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Label constrained convolutional factor analysis for classification with limited training samples

Jian Chen, Lan Du *, Yuchen Guo

National Lab of Radar Signal Processing, Xidian University, Xi'an 710071, China

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ABSTRACT

This paper mainly addresses the statistical classification robust to small training data size. We develop a label constrained convolutional factor analysis (LCCFA) model, which unifies the factor analysis (FA), convolution operation and supervised learning. In the LCCFA model, each dictionary atom is used as a small-sized convolution kernel with the goal of learning the observations' basic structures, which have highly shared characteristics among all observed data. This property enables the proposed method to describe data with fewer dictionary atoms than the FA model and reduces the model complexity. Consequently, the classification performance of the LCCFA model can be improved in the case of limited training samples. Meanwhile, the proposed model also projects the weight vectors of dictionary atoms to their class labels to constrain the learning of parameters. The difference in weight vectors from different classes increases due to the label constraint, thereby offering the potential to enhance the inter-class separability of statistical models. Additionally, the efficient parameter estimation is implemented via variational Bayesian (VB) algorithm. Experimental results on several benchmark datasets and measured radar high-resolution range profile (HRRP) data show that our method outperforms other related models in terms of small sample classification.

1. Introduction

In many fields, such as biomedicine, artificial intelligence, and remote sensing, automatic classification has become an important technology and been widely studied by researchers. There is a broad consensus on the fact that numerous training samples are required for a classification model to capture the full description of objects, ultimately leading to the satisfactory classification performance [24]. Nevertheless, due to the difficulty in data collection and high cost of data annotation, there are not enough training samples for medicine, aerospace, natural language processing, and other specific areas. In the case of limited training samples, a model often suffers from the so-called "small sample size" problem [34], such as weak classification ability and poor generalization performance. Therefore, it is of great practical significance to study the classification method with limited samples, and small sample classification technology is becoming a major research topic.

The existing methods for classification with small training data size can be divided into three main categories: transfer learning-based method [4,8,16,27,39,40,45,46], data augmentation-based method [3,14,36,43], and Bayesian probability model-based method [9,12,13,22,29,35,37,41,42]. Although the first two approaches mitigate the small sample problem to a certain extent, the transfer learning technology heavily depends on the similarity between the auxiliary data and

* Corresponding author at: 2 Taibai Rd., Xi'an, Shaanxi 710071, China.
E-mail address: dulan@mail.xidian.edu.cn (L. Du).

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observed data, and data augmentation is substantially affected by the quality of the generated data, thus limiting the wide applicability of these two methods. As an alternative way, the Bayesian probability model-based method builds a parametric model to describe observations and has three main advantages. (1) Prior knowledge can be integrated into the model construction. According to the Bayesian theory, a proper prior is beneficial for the case of small data learning. (2) The probability distribution of the model parameter can be estimated, which provides parameter's uncertainty measure. Therefore, compared to the model constructed in the algebraic framework where the parameter point estimation is performed, the Bayesian probability model can acquire the better generalization capacity when there are limited training samples. (3) Lastly, nonparametric Bayesian theory can be exploited to adaptively determine the capacity of the model according to the size of training data, avoiding the over-fitting under the condition of small samples. Due to these superiorities, small sample classification based on Bayesian probability model has attracted extensive attentions.

In [13], the principal component analysis (PCA)-based subspace approximation model and probabilistic PCA (PPCA) model are developed for radar target recognition. The factor analysis (FA) model, which is a probabilistic latent feature model, is proposed in [13,37]. In particular, FA is able to reveal underlying information in observed data and extensively used in pattern classification. Du et al. [13] utilize FA model to describe the statistical correlations among different range cells of a radar high-resolution range profile (HRRP) sample. Tunc et al. [37] propose a class dependent FA (CDFA) model by combining the FA model with manifold learning and apply the model to face classification. In [35], an improved FA model, called local FA (LFA), is developed to describe the non-Gaussian characteristic of the data distribution. To obtain an enhanced classification performance in the case of limited training samples, some literatures [12,29,42] propose to share some parameters among different Bayesian probability models by referring to the thought of information sharing in the multi-task learning (MTL), thereby decreasing the number of parameters to be estimated and improving the classification accuracy under small training data size condition. In [42], the whole dataset is divided into several clusters via the Dirichlet process (DP) mixture and the observed data in each cluster are classified by using the corresponding logistic regression classifier with all classifiers being connected via a common prior on the classifier parameters. In our recent work [12], an MTL-FA model is proposed, which automatically selects a sub-projection matrix for each model from a shared loading matrix among all observed samples, thus exhibiting the lower model complexity than the traditional FA model.

The preceding models mainly focus on the acquisition of observations' global information. Recently, many convolutional methods, including convolutional restricted Boltzmann machines (RBMs) [25,30], deconvolutional networks [28,31], and convolutional neural networks (CNNs) [1,15,26,44], have attracted a lot of attentions. For these methods, convolution kernels are capable of extracting observations' local details and play an important role in the reduction of network parameters and complexity. Inspired by this, the convolution operation can also be introduced into the MTL-FA model to further decrease the demand on training sample size. In our method, each dictionary is treated as a convolution kernel, and the observed sample is modeled as the convolution summation of several dictionary atoms and their weight vectors. In this way, the dictionary atom has the local receptive field on observations, leading to the acquisition of essential information behind the observed data. Therefore, our method should be able to depict data by using the smaller number of dictionary atoms than the MTL-FA model. Because of the smaller dictionary size, the proposed model has fewer estimating parameters and can perform better when the training data we got are insufficient.

So far the methods we have presented are unsupervised. That is, they only focus on observations and do not utilize label information. However, in classification task where labels of data are available, it would be more helpful to incorporate supervised information into the statistical modeling. Besides in the view of model complexity reduction, another contribution of this paper is that we extend our convolutional FA to a supervised model, i.e., label constrained convolutional FA (LCCFA) model, with the introduction of additional information, i.e., sample labels. In the LCCFA model, the class labels of observed data are also used to restrict the learning of the model parameters, which makes statistical characteristics of models from different classes more distinguishable. Therefore, the proposed method not only weakens the sensitivity of the model to the number of training samples, but also enhances the discriminability of the statistical model. These two aspects lead to the greater improvement in classification ability of the model with small set of samples.

The research reported here mainly concentrates on the Bayesian probability model-based small sample classification and proposes a LCCFA model on the basis of MTL-FA. We summarize the contributions of the proposed method as follows:

- Enlightened by the convolutional networks, the convolution operation is introduced in the LCCFA model, which further reduces the number of model parameters to be estimated and reinforces the model advantage in low complexity. Therefore, our LCCFA model is more applicable to the small sample classification than the MTL-FA model.
- To enhance the discriminant power of the statistical model, the proposed method also makes full use of the label information of data, with which the differences in statistical characteristic of models for different classes can be much larger, thus further improving the classification performance of the model with limited training samples.

As discussed above, the current approaches in small sample classification based on Bayesian probability model mainly resort to the model construction with the fewer parameters. In this regard, our work provides a novel modeling method that has the lower complexity and can be selected as an alternative method to deal with the classification with small training data size. Besides the perspective of the reduction in model parameters, the label constraint adopted in our work also offers a new idea for the current small sample classification techniques, i.e., the use of other available information. With the help of addition information, the existing methods should also have great potentiality for classification improvement in the case of limited training samples.

Table 1 lists all notations in our LCCFA model to assist readers clearly understand the matrix dimensions and definitions of notations.

The remainder of the paper is organized as follows. Section 2 discusses the methods with respect to small sample classification. Section 3.1 reviews FA models, including STL-FA and MTL-FA, and Section 3.2 then introduces the LCCFA model in detail. Sections 4.1 and 4.2 provide the model reasoning and statistical classification scheme based on the LCCFA model, respectively. Section 5 conducts experiments on some benchmark data sets and measured HRRP data sets. Finally, Section 6 summarizes the paper.

2. Related work

As discussed in Introduction, works on small sample classification are carried out mainly from transfer learning, data augmentation, and Bayesian probability modeling. This section will briefly introduce research in these three fields to position our method within the current literatures.

2.1. Small sample classification with transfer learning

In this method, mass data from another domain (called source domain) are exploited to assist the model learning in the current domain with insufficient training samples (called target domain). The transfer learning (TL) approach can be summarized into two cases. The first case is the fine-tuning based TL [8,45], in which a base model is trained with source data, and then some model parameters are directly reused to conduct a target model with its remaining parameters being randomly initialized. During the training of the target model, the transferred parameters from the source model can be fine-tuned or frozen. The second case is the domain adaptation (DA) based TL [39]. The DA method tends to find a feature space where the latent representations of the source and target data are similar so that the classifier trained with the source data features can also be applicable to those of the target data. References [27] and [40] learn the transferable features that bridge the cross-domain difference with the minimization of maximum mean discrepancy (MMD), and the same end is achieved via the Kullback-Leibler (KL) divergency constraint in [46]. In [16], a domain classifier is introduced to identify whether the input feature is from source domain or from target domain. By minimizing the confusion loss of the domain classifier, the feature distributions over two domains tend to be similar. Bousmalis et al. [4] propose a domain separation network, where a shared-weight model learns to capture the shared representations from different data domains, while a private model is used for domain-specific components in each domain. By partitioning the space in such a manner, the shared representations will not be influenced by domain-specific representations. Accordingly, a good transfer ability can be obtained.

Although the TL technique mitigates the small sample problem, the classification performance of these methods heavily depends on the data similarity between the source and target domains. Usually, the lower the similarity degree is, the poorer the classification performance is. However, it is hard to find the appropriate source data for some classification tasks, such as the radar target recognition due to its special data formation mechanism.

2.2. Small sample classification with data augmentation

This method focuses on generating some synthetic samples with great similarity to observations. Thereafter, the generated data are incorporated with the existing data to train the classifier. In [14], data augmentation operations, including translation, speckle noising, and pose synthesis, are adopted to enlarge synthetic aperture radar (SAR) images for radar target recognition. Apart from the manual data expansion, generative networks, such as variational autoencoder (VAE) [23] and

Table 1
List of notations in the LCCFA model.

Notations	Meanings	Notations	Meanings
C	The number of classes	$\mathbf{w}_{nk} \in \mathbb{R}^L$	Weight vector corresponding to $\mathbf{d}_k$ for $\mathbf{x}_n^c$, with its ith element being w_{nki}
N_c	The number of samples from the cth class	$\mathbf{d}_{ki} \in \mathbb{R}^P$	The ith shifted version of $\mathbf{d}_k$
P	Dimension of the observed sample	$\mathbf{A} \in \mathbb{R}^{C\times KL}$	Label projection matrix with its column $\mathbf{a}_{(k-1)L+i} \in \mathbb{R}^C$
K	The number of dictionary atoms	$\varepsilon_n^c \in \mathbb{R}^P$	Noise variable of $\mathbf{x}_n^c$ with the covariance precision beingγ^c
J	Dimension of the dictionary atom	$\eta_n^c \in \mathbb{R}^C$	Noise variable of $\mathbf{q}_n^c$ with the covariance precision being δ^c
L	Dimension of the weight vector with$L = P - J + 1$	b_{nk}^c	The dictionary atom selector of $\mathbf{x}_n^c$ with$k = 1, \cdots, K$
$\mathbf{x}_n^c \in \mathbb{R}^P$	The nth observed sample from the cth class	α_{nki}^c	The inverse variance ofw_{nki}
$\mathbf{q}_n^c \in \mathbb{R}^C$	Class label of$\mathbf{x}_n^c$	β_{kj}	The inverse variance ofd_{kj}
$\mu_n^c \in \mathbb{R}^P$	Mean vector of samples from the cth class	π_k^c	Prior that controls the sparsity of b_{nk}^c
$\mathbf{d}_k \in \mathbb{R}^J$	The kth shared dictionary atom with its jth element being d_{kj}	$a_0 \cdots h_0, p_0, q_0, v_0, u_0$	Hyperparameters of the model

generative adversarial network (GAN) [17], are also exploited to perform data augmentation. Samuel et al. [36] and Xu at al. [43] respectively use recurrent neural network (RNN)-based and long short-term memory (LSTM) network-based VAEs to generate sentences for text classification. In [3], synthetic images are produced via the GAN, in which the discriminator assures synthetic images generated by the generator similar to true images and an additional classifier is also trained to predict the class labels of all images.

Above data augmentation methods increase the amount of data via the generation of synthetic samples, thus alleviating the small sample problem to a certain extent. However, the quality of the generated data has a significant effect on the classification performance, and furthermore, it is difficult to choose proper samples from the synthetic data for classifier training. These drawbacks limit the wide applicability of data augmentation-based small sample classification.

2.3. Small sample classification with Bayesian probability model

Due to the superiorities of probabilistic models demonstrated in Introduction, statistical modeling method is also selected as an alternative way to solve the classification with small training samples. In [9,41,22], adaptive Gaussian classifier (AGC), linear dynamic model, and Gamma mixture model are proposed for radar target recognition, all of which have the fewer model parameters and less demand for the training samples. However, since they cannot comprehensively describe the statistical characteristics of data, their overall recognition performance is not so good in spite of the high recognition rate in the case of only a few training samples. In this regard, the researchers propose PCA-based subspace approximation [13], PPCA [13], and FA [13,37] models. These models provide a fuller description of the observed data and boost the overall performance at the expense of additional model complexity, thus obtaining the lower classification accuracy than the aforementioned probability models when the training data are not enough. For the sake of stronger classification ability under the small sample circumstance while assuring the overall performance of the model, the information sharing-based models [12,29,42] are proposed, especially the MTL-FA model because of its strong ability to reveal the latent representations of observations with a small amount of training data, as discussed in Introduction.

Our formulation for small sample classification is also from the aspect of Bayesian probability modeling and related to the MTL-FA model. We propose a LCCFA model, which further reduces the model parameters to be estimated compared with the MTL-FA by introducing the convolution operation. Besides in the view of model complexity reduction, we also exploit the additional information, i.e., sample labels, to adjust the learning of parameters, thereby enabling the models from different classes have considerable separability. These two advantages of the LCCFA model make it possible to obtain greater performance improvement in small sample classification.

3. Label constrained convolutional factor analysis model

3.1. Statistical modeling based on the FA model

FA describes a linear generation process from the latent variable distribution to data distribution. For a dataset with C classes and each class containing N_c $(c = 1, \cdots, C)$ samples, the generative FA models the observation by $\mathbf{x}_n^c = \mathbf{D}^c\mathbf{w}_n^c + \mu^c + \varepsilon_n^c$, where $\mathbf{x}_n^c \in \mathbb{R}^P$ is the nth $(n = 1, \cdots, N_c)$ observed sample from the cth class with its class statistical mean being $\mu^c \in \mathbb{R}^P$, $\mathbf{D}^c$ is the loading matrix of size $P \times V$. In addition, the latent factor variable $\mathbf{w}_n^c \in \mathbb{R}^V$ is projected to $\mathbf{x}_n^c$ by $\mathbf{D}^c$ and usually assigns a isotropic Gaussian prior, i.e., $\mathbf{w}_n^c \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_V)$. $\varepsilon_n^c \in \mathbb{R}^P \sim \mathcal{N}(\mathbf{0}, \Psi^c)$ is the model noise with Ψ^c being a diagonal matrix. Then we can obtain the marginal density function $p(\mathbf{x}_n^c) = \mathcal{N}\left(\mathbf{x}_n^c \middle| \mu^c, \mathbf{D}^c\mathbf{D}^{cT} + \Psi^c\right)$ according to the characteristic of Gaussian distribution. During the training phase, each class learns a FA model and the model learning of all classes is irrelevant. We refer to above modeling approach as the single-task learning (STL) based FA, i.e., STL-FA model.

Reference [12] shows that the reliable parameter estimation of the STL-FA model can be achieved in the case of abundant training samples. However, sufficient training data are not constantly available in many applications. For example, a small sample size problem often occurs in radar target recognition because of the contradiction between the high velocity of target and limited sampling rate of radar [12,13]. As a result, the parametric model has lower estimation accuracy. Our previous work [12] presents an MTL-based modeling method, with which the FA models of all classes are learned jointly by sharing the loading matrix across all training samples, i.e., MTL-FA. By information sharing, the MTL-FA model needs to estimate fewer parameters than the STL-FA model, which alleviates the problem of small sample learning in statistical classification. The MTL-FA model can be concretely expressed as

$$\mathbf{x}_n^c = \mathbf{D}\left(\mathbf{w}_n^c \odot \mathbf{z}_n^c\right) + \mu^c + \varepsilon_n^c$$

$$\begin{aligned} \mathbf{d}_v &\sim \mathcal{N}(0, 1/P\mathbf{I}_P), \\ \mathbf{w}_n^c &\sim \mathcal{N}(0,\ 1/\alpha^c\mathbf{I}_V),\ \ \alpha^c \sim Gamma(a_0,\ b_0), \\ z_{n,v}^c &\sim Bernoulli(u_v^c),\ \ u_v^c \sim Beta(c_0,\ d_0), \\ \varepsilon_n^c &\sim \mathcal{N}(0,\ 1/\gamma^c\,\mathbf{I}_P),\ \ \gamma^c \sim Gamma(e_0,\ f_0), \end{aligned} \tag{1}$$

where $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \cdots, \mathbf{d}_V] \in \mathbb{R}^{P \times V}$ is the shared loading matrix with $\mathbf{d}_v$ for $v = 1, \cdots, V$ denoting the vth column of $\mathbf{D}$ and V being set to a large value. Due to the over-completeness of loading matrix $\mathbf{D}$, only a few columns in $\mathbf{D}$ are used to generate the observation $\mathbf{x}_n^c$. For the column selection, a binary vector $\mathbf{z}_n^c = \left[z_{n,1}^c, \cdots, z_{n,v}^c, \cdots, z_{n,V}^c\right]^T$ is appended to the latent variable $\mathbf{w}_n^c$ with $\odot$ representing the point-wise (Hadamard) vector product. When the value of $z_{n,v}^c$ is 1, the vth column contributes to the representation of $\mathbf{x}_n^c$, and the opposite meaning for $z_{n,v}^c$ being 0. $\mu^c \in \mathbb{R}^P$ is the mean vector, and $\varepsilon_n^c \in \mathbb{R}^P$ is the noise variable. The Gamma density $Gamma(\cdot)$, Bernoulli density $Bernoulli(\cdot)$, and Beta density $Beta(\cdot)$ act as the super-prior distributions with their hyperparameters being a_0, b_0, c_0, d_0, e_0, and f_0, to control the sparsity of the corresponding model parameters.

In contrast with the STL-FA model where V is preset and fixed, the MTL-FA model can automatically determine V via the Beta process (BP) construction [12,32], which avoids over-fitting or under-fitting caused by excessively high or low value of V. In addition, the model in (1) imposes the hierarchical prior on the parameter. From the view of Bayes, it is beneficial for the more accurate model learning by introducing more and appropriate prior knowledge in the context of insufficient training data. Actually, FA is the Bayesian form of dictionary learning, where the loading matrix $\mathbf{D}$ can be seen as the dictionary. The column in $\mathbf{D}$ and latent variable $\mathbf{w}_n^c$ can further be referred to as the dictionary atom and weight vector of $\mathbf{x}_n^c$ under $\mathbf{D}$, respectively.

3.2. LCCFA model construction

As shown in (1), the MTL-FA model considers the observed sample as the weighted summation of bases with their dimensions equal to that of the sample. This modeling approach describes the observation's generative process from the global perspective. However, many recent studies based on convolutional networks tend to describe observations from the idea of local perception because only spatially nearby elements are highly correlated in a sample, i.e., the local correlation [26], which goes back to Hubel and Wiesel's discovery of locally sensitive neurons in the cat's visual system [20]. In combination with the local correlation, we can also consider that the observed sample is constituted by several local components. Distinct from the FA model where the dictionary atom (i.e., the basis vector) extracts the global information, we hope each dictionary atom in the proposed model is capable of capturing the local information of the observed sample and different dictionary atoms can capture diverse local information. Ultimately, multiple dictionary atoms together make up a complete sample.

In addition, the local information extracted by a dictionary atom is likely to make available for different local regions of a sample. This is because the local detail can better reflect the nature of the observation, thus having the stronger sharing ability in different parts of an observed sample than the global information. With this regard, the sharing property of local information in a sample should also be considered during model construction from the view of local perception.

Combined with the above analyses, the convolution of the dictionary atom and the corresponding weigh vector can realize the generation of data from the aspects of local perception and sharing of the dictionary atom in a sample. On the one hand, the small-scaled dictionary atom can learn the local information of data. On the other hand, each dictionary atom acts as a convolution kernel and slides on the corresponding weigh vector whose dimension is similar to that of data. When the local information described by a dictionary atom appears in one part of the observation, the weight in the corresponding position of the weight vector is large; otherwise, the weight is small. Due to the sliding characteristics of the convolution, the information learned by a dictionary atom is considered in all local regions of a sample, i.e., the sharing of the dictionary atom in a sample. Lastly, the convolutional summation of different dictionary atoms and their weight vectors assures the generation of a complete sample.

Therefore, we introduce the convolution operation to the MTL-FA model and represent the observation $\mathbf{x}_n^c$ as

$$\mathbf{x}_n^c = \sum_{k=1}^{K} b_{nk}^c \mathbf{d}_k * \mathbf{w}_{nk}^c + \mu^c + \varepsilon_n^c \tag{2}$$

where $*$ is the convolution operator; $\mu^c \in \mathbb{R}^P$ is the mean variable and can be directly expressed as $\mu^c = \frac{1}{N}\sum_{n=1}^{N} \mathbf{x}_n^c$; $\mathbf{d}_k \in \mathbb{R}^J$ ($k = 1, \cdots, K$ with K being initialized to a large value) is the kth dictionary atom, and its size is much smaller than that in the MTL-FA, i.e., $J \ll P$; $b_{nk}^c \in \{0, 1\}$ indicates whether $\mathbf{d}_k$ is used to represent $\mathbf{x}_n^c$; $\varepsilon_n^c \in \mathbb{R}^P$ represents the noise variable, and $\mathbf{w}_{nk}^c \in \mathbb{R}^L$ is the weight vector corresponding to $\mathbf{d}_k$ for observed data $\mathbf{x}_n^c$. As we see from (2), the LCCFA treats the J-dimensional dictionary atom $\mathbf{d}_k$ ($k = 1, \cdots, K$) as the convolution kernel of the corresponding L-dimensional weight vector $\mathbf{w}_{nk}^c$ and slides $\mathbf{d}_k$ on $\mathbf{w}_{nk}^c$ to generate the P-dimensional $\mathbf{x}_n^c$. According to the property of linear convolution (full convolution rather than the valid convolution), the result of $\mathbf{d}_k * \mathbf{w}_{nk}^c$ is of size $L + J - 1$. Thus, we have $P = L + J - 1$, and then $L = P - J + 1$.

In contrast with the basis vector in the MTL-FA model, each dictionary atom in (2) plays the role of convolution kernel with small size and is prone to capture local and basic structure of observations. The strong sharability of the local feature makes a good representation of all observed samples possible with only a few dictionary atoms. Therefore, the construction in (2) has a small number of model parameters and is considerably suitable for small sample classification. However, the above modeling approach is unsupervised. Actually, the class label is the useful information for classification. With this consideration, we propose a label constrained convolutional FA model, i.e., LCCFA model, which takes advantage of class label to restrict the learning of model parameters while depicting the observed data with the convolutional FA. The introduction of label information enhances the separability of statistical models among different classes. Furthermore, the prediction performance of our LCCFA model can be improved.

We suppose that a binary vector $\mathbf{q}_n^c = [0, \cdots, 0, 1, 0, \cdots, 0]^T \in \mathbb{R}^C$ is the class label corresponding to $\mathbf{x}_n^c$, and the nonzero value of $\mathbf{q}_n^c$ only occurs at the cth index. In our LCCFA model, the observed data $(\mathbf{x}_n^c, \mathbf{q}_n^c)$ is generated as follows:

$$\begin{aligned} \mathbf{x}_n^c &= \sum_{k=1}^{K} b_{nk}^c \mathbf{d}_k * \mathbf{w}_{nk}^c + \mu^c + \varepsilon_n^c \\ \mathbf{q}_n^c &= \mathbf{A}\hat{\mathbf{w}}_n^c + \eta_n^c \\ \hat{\mathbf{w}}_n^c &= \left[b_{n1}^c \mathbf{w}_{n1}^{c\mathrm{T}},\ b_{n2}^c \mathbf{w}_{n2}^{c\mathrm{T}},\ \cdots,\ b_{nK}^c \mathbf{w}_{nK}^{c\mathrm{T}}\right]^{\mathrm{T}} \end{aligned} \tag{3}$$

where $\mathbf{A} = [\mathbf{A}_1, \cdots, \mathbf{A}_k, \cdots, \mathbf{A}_K] \in \mathbb{R}^{C\times KL}$ is the projection matrix with $\mathbf{A}_k = \left[\mathbf{a}_{(k-1)L+1},\ \mathbf{a}_{(k-1)L+2}\ , \cdots, \mathbf{a}_{(k-1)L+L}\right] \in \mathbb{R}^{C\times L}$ and η_n^c is the noise variable. It should be pointed that the proposed method in (3) is over-complete because all weight vectors $\{\mathbf{w}_{nk}^c\}_{k=1,\cdots,K}$ have a substantially larger size than the observed data $\mathbf{x}_n^c$, i.e., $K(P-J+1) \gg P$. This situation results in the dictionary's non-unique solutions, some of which are unable to extract the structural information of observations. To deal with this problem, the max-pooling technique is adopted after learning each weight vector. In detail, the maximum value in the pooling area remains unchanged and the others are set to zero. Some studies [25,30] have taken a similar approach to achieve structured dictionary atoms.

The LCCFA model can be rewritten as

$$\begin{aligned} \mathbf{x}_n^c &= \sum_{k=1}^{K}\sum_{i=1}^{L} b_{nk}^c w_{nki}^c \mathbf{d}_{ki} + \mu^c + \varepsilon_n^c \\ \mathbf{q}_n^c &= \sum_{k=1}^{K}\sum_{i=1}^{L} b_{nk}^c w_{nki}^c \mathbf{a}_{(k-1)L+i} + \eta_n^c \end{aligned} \tag{4}$$

where $w_{nki}^c\ (i = 1, \cdots, L)$ represents the ith element of $\mathbf{w}_{nk}^c$. Moreover, $\mathbf{d}_{ki} \in \mathbb{R}^P$ is generated in the following way: firstly, zero padding is operated at the end of $\mathbf{d}_k$ to construct a new vector of size $P \times 1$; secondly, a circular shift by $i-1$ units is implemented for the zero-padded vector to obtain $\mathbf{d}_{ki}$. Similar to the MTL-FA, the hierarchical prior setting is exploited in the LCCFA model:

$$\begin{aligned} w_{nki}^c &\sim \mathcal{N}(0, 1/\alpha_{nki}^c), \alpha_{nki}^c \sim Gamma(a_0, b_0), \\ b_{nk}^c &\sim Bernoulli(\pi kc), \pi kc \sim Beta(c_0, d_0), \\ d_{kj} &\sim \mathcal{N}(0, 1/\beta_{kj}), \beta_{kj} \sim Gamma(e_0, f_0), j = 1, \cdots, J, \\ \mathbf{a}_{(k-1)L+i} &\sim \mathcal{N}(\mathbf{0},\ 1/\lambda_{ki}\mathbf{I}_C), \lambda_{ki} \sim Gamma(g_0, h_0), \\ \varepsilon_n^c &\sim \mathcal{N}(\mathbf{0},\ 1/\gamma^c\mathbf{I}_P), \gamma^c \sim Gamma(p_0, q_0) \\ \eta_n^c &\sim \mathcal{N}(\mathbf{0},\ 1/\delta^c\mathbf{I}_c), \delta^c \sim Gamma(u_0, v_0), \end{aligned} \tag{5}$$

where J is the dictionary atom dimension; d_{kj} represents the jth element in $\mathbf{d}_k$; $a_0, b_0, c_0, d_0, e_0, f_0, g_0, h_0, p_0, q_0, u_0$, and v_0 are the preset hyperparameters. For intuitive representation of data generation, we give the graphical model of LCCFA in Fig. 1.

From (3) and (5), we see that the weight vectors $\{\mathbf{w}_{nk}^c\}_{k=1,\cdots,K}$ can be regarded as the "bridge" of the observed sample $\mathbf{x}_n^c$ and class label $\mathbf{q}_n^c$. Thus, the estimation of $\{\mathbf{w}_{nk}^c\}_{k=1,\cdots,K}$ will be affected by $\mathbf{x}_n^c$ and $\mathbf{q}_n^c$. In this study, the posteriors of the model parameters are estimated via variational Bayesian (VB) algorithm [5]. Based on Bayes theorem, we can express the posterior distribution of $\mathbf{w}_{nk}^c$, i.e., $p(\mathbf{w}_{nk}^c|\mathbf{x}_n^c, \mathbf{q}_n^c)$, as follows:

$$p(\mathbf{w}_{nk}^c|\mathbf{x}_n^c, \mathbf{q}_n^c) \propto p(\mathbf{x}_n^c|\mathbf{w}_{nk}^c) p(\mathbf{q}_n^c|\mathbf{w}_{nk}^c) p(\mathbf{w}_{nk}^c) \tag{6}$$

For $p(\mathbf{x}_n^c|\mathbf{w}_{nk}^c)$ given other model parameters, including $\mathbf{D}$, $\mathbf{A}$, $\{b_{nk}^c\}_{k=1,\ldots,K}$, $\{\mathbf{w}_{nk}^c\}_{k=1,\cdots,k-1,k+1,\cdots,K}$, $\mu^{c,m}$, and the prior distribution $\varepsilon_n^c \sim \mathcal{N}(\mathbf{0},\ 1/\gamma^c\mathbf{I}_P)$, we have

$$p(\boldsymbol{x}_n^c|\mathbf{w}_{nk}^c) = (2\pi)^{-\frac{P}{2}}|1/\gamma^c\boldsymbol{I}_P|^{-\frac{1}{2}}\exp\left(-\frac{\gamma^c}{2}\boldsymbol{x}_{-n}^{cT}\boldsymbol{x}_{-n}^c\right) \tag{7}$$

where $\mathbf{x}_{-n}^c = \mathbf{x}_n^c - \mu^c - \sum_{k=1}^{K} b_{nk}^c \mathbf{d}_k * \mathbf{w}_{nk}^c$. Similarly for $p(\mathbf{q}_n^c|\mathbf{w}_{nk}^c)$,

$$p(\boldsymbol{q}_n^c|\boldsymbol{w}_{nk}^c) = (2\pi)^{-\frac{C}{2}}|1/\delta^c\boldsymbol{I}_C|^{-\frac{1}{2}}\exp\left(-\frac{\delta^c}{2}\boldsymbol{q}_{-n}^{cT}\boldsymbol{q}_{-n}^c\right) \tag{8}$$

where $\mathbf{q}_{-n}^c = \mathbf{q}_n^c - \mathbf{A}\hat{\mathbf{w}}_n^c$. In combination of (6), (7), and (8), we know that the weight vector $\mathbf{w}_{nk}^c$ is not only related to the observation $\mathbf{x}_n^c$, but also is affected by the corresponding class label $\mathbf{q}_n^c$. Due to the distinctions of class labels, the weight vec-

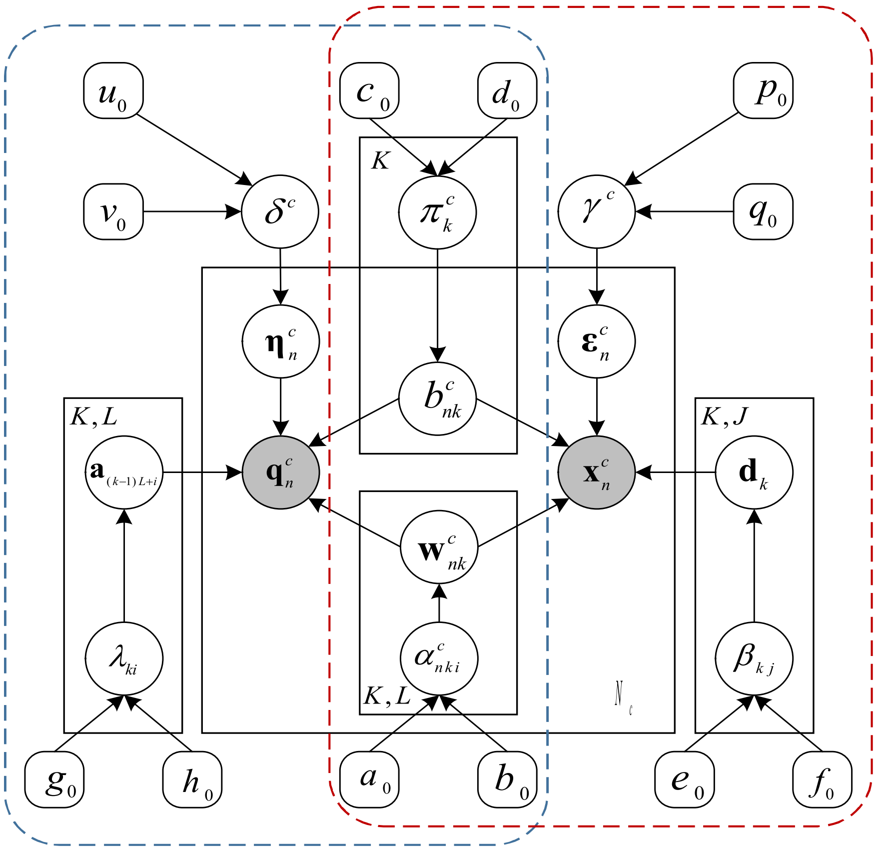

Fig. 1. Graphical representation of the LCCFA model. The right part framed with the red dotted line represents the convolutional FA module, and the left part framed with the blue dotted line represents the label constraint structure. The notations in the corner of the black solid boxes, including N_c, K, J, and L, represent the boxes' stack for indexes n, k, j, and i, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

tors from different classes in our LCCFA model should have greater differences than those in (2). Accordingly, the LCCFA model has the stronger discriminative capability while extracting the observations' structure information.

In summary, the proposed LCCFA model incorporates convolution operation with discriminative information. On the one hand, the introduction of convolution operation enables the modeling of data with the lower freedom. On the other hand, the label constraint promotes the model learning towards the direction of more separable. Thus, the classification capability of the proposed LCCFA model can be further improved under limited training data conditions.

4. Statistical classification with the LCCFA model

In our work, we build a LCCFA model for the description of observations. With the LCCFA model, a Bayes classifier can be designed and used thereafter to the statistical classification of the unknown data [21]. For a clear illustration, we give the flowchart of the statistical classification based on the LCCFA model in Fig. 2.

As shown in Fig. 2, the integrated scheme includes two stages, i.e., training stage in the solid line frame and classification stage in the dotted line frame. Then we introduce the two stages in detail.

- Training stage

In this stage, the observed data are firstly required to be preprocessed, such as normalization operation. Thereafter, the LCCFA model is built among all observations from all classes, and the model learning, i.e., parameter estimation, can be subsequently performed. Once the model learning completes, a Bayes classifier can be designed for the statistical classification of the test sample.

The essence of the Bayes classifier is to estimate the class posterior probability density function (PDF) of data from each class via the learned parameters in the corresponding class, i.e., $p(c|\mathbf{x})$ with $\mathbf{x}$ denoting the observed data, $c = 1, \cdots, C$

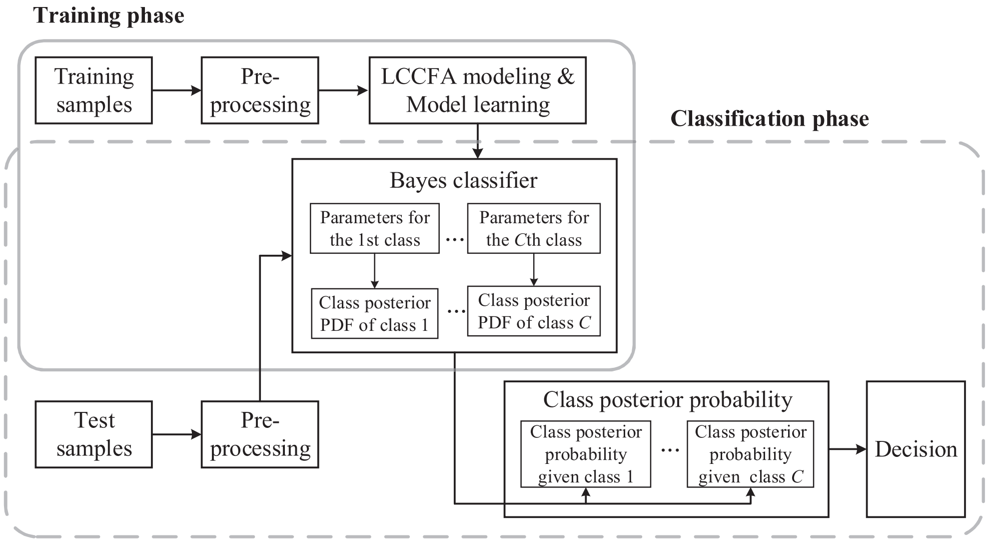

Fig. 2. Flowchart of the statistical classification based on the LCCFA model.

denoting the class index and C denoting the number of classes. Then the Bayes classifier determines the class membership of a test sample $\mathbf{x}^*$ according to its class posterior probabilities under all classes, i.e., $\{p(c|\mathbf{x}^*)\}_{c=1}^{C}$.

- Classification stage

In this stage, after data preprocessing, the test sample is fed into the constructed Bayes classifier, with which the class posterior probabilities of the test sample given all classes can be calculated. Ultimately, this unknown sample is assigned to the class with the maximum class posterior probability.

In the following, we give a detailed introduction to the model learning of the LCCFA model and provide an estimation method for the class posterior PDF used in the Bayes classifier.

4.1. Model learning exploiting VB algorithm

Usually, Gibbs sampling-based Markov chain Monte Carlo method [18] and VB algorithms are two common methods used to estimate the model parameters. In this paper, we adopt VB inference due to its higher computational efficiency.

VB inference aims to obtain the approximate estimation of the model parameters' true posteriors [2]. It can be expressed mathematically as

$$\min_{q(\Phi)} \; KL(q(\Phi)\| \, p(\Phi|\mathbf{X},\theta)) \tag{9}$$

where $q(\Phi)$ is the variational distribution with Φ representing the model parameters, $p(\Phi|\mathbf{X},\ \theta)$ is the true posterior of Φ with $\mathbf{X}$ representing the observations and θ the hyperparameters. $KL(\cdot\|\ \cdot)$ is the Kullback-Leibler (KL) divergence that measures the closeness between two distributions. Especially, the KL divergency in (9) is 0 when $q(\Phi) = p(\Phi|\mathbf{X},\theta)$.

We can further expand the KL divergency in (9) as follows:

$$\begin{aligned} KL(q(\Phi)\|\, p(\Phi|\mathbf{X},\theta)) &= \int q(\Phi)\ln\tfrac{q(\Phi)}{p(\Phi|\mathbf{X},\theta)}d\Phi \;= \int q(\Phi)\ln\tfrac{q(\Phi)p(\mathbf{X}|\theta)}{p(\mathbf{X},\Phi|\theta)}d\Phi \\ &= \int q(\Phi)\ln p(\mathbf{X}|\theta)d\Phi - \int q(\Phi)\ln\tfrac{p(\mathbf{X},\Phi|\theta)}{q(\Phi)}d\Phi \\ &= \ln p(\mathbf{X}|\theta) - LB(q(\Phi)) \end{aligned} \tag{10}$$

where $\ln p(\mathbf{X}|\theta)$ is the log marginal likelihood of $\mathbf{X}$ and $LB(q(\Phi))$ is the lower bound of $\ln p(\mathbf{X}|\theta)$. The marginal likelihood $\ln p(\mathbf{X}|\theta)$ is a constant with respect to $q(\Phi)$. Thus, minimizing the KL divergency in (9) is turned into maximizing the corresponding lower bound.

According to mean field-based VB theory [33], we often assume that the posteriors of the model parameters are mutually independent, i.e.,

$$q(\Phi) = \prod\nolimits_k q_k(\varphi) \tag{11}$$

where $q_k(\varphi)$ represents the posterior distribution of each model parameter. This assumption guarantees the tractability of the lower bound's computation. In the iterative solution of $q_k(\varphi)$, the value of the lower bound $LB(q(\Phi))$ increases until convergence.

In the LCCFA model, $\mathbf{X} = \{\mathbf{x}_n^c\}_{n=1,c=1}^{N_c,\ C}$ is the observed dataset, $\theta = \{a_0, b_0, c_0, d_0, e_0, f_0, g_0, h_0, p_0, q_0, u_0, v_0\}$ is the collection of hyperparameters, and $\Phi = \{b_{nk}^c, \pi_k^c, \mathbf{d}_k, \mathbf{a}_{(k-1)L+i}, \mathbf{w}_{nk}^c, \alpha_{nk}^c, \beta_k, \lambda_{ki}, \gamma^c, \delta^c\}_{i=1,k=1,n=1,c=1}^{L,\ K,\ N_c,\ C}$ with $\alpha_{nk}^c = [\alpha_{nk1}^c, \cdots, \alpha_{nkL}^c]^T$ and $\beta_k = [\beta_{k1}, \cdots, \beta_{kJ}]^T$ is the parameter set of the model. We summarize the appropriate posteriors of each parameters as follows.

1) For α_{nk}^c: we assume $q(\alpha_{nki}^c) = Gamma\left(\alpha_{nki}^c \middle| \tilde{a}_{nki}^c, \tilde{b}_{nki}^c\right)$, and the distribution parameters are calculated as

$$\begin{aligned} \tilde{a}_{nki}^c &= a_0 + \tfrac{1}{2} \\ \tilde{b}_{nki}^c &= b_0 + \tfrac{1}{2}\left\langle \left(w_{nki}^c\right)^2 \right\rangle \end{aligned} \tag{12}$$

with symbol indicating the expectation of the posterior.

2) For $\mathbf{w}_{nk}^c$: we assume $q(w_{nki}^c) = \mathcal{N}(w_{nki}^c | \varsigma_{nki}^c, \Sigma_{nki}^c)$, and the updating formulas are

$$\begin{aligned} \Sigma_{nki}^c &= \left(\langle \alpha_{nki}^c \rangle + \langle \gamma^c \rangle \langle b_{nk}^c \rangle \left\langle \| \mathbf{d}_k \|_2^2 \right\rangle + \langle \delta^c \rangle \langle b_{nk}^c \rangle \left\langle \| \mathbf{a}_{(k-1)L+i} \|_2^2 \right\rangle \right)^{-1} \\ \varsigma_{nki}^c &= \Sigma_{nki}^c \langle b_{nk}^c \rangle \Big[\langle \gamma^c \rangle \left(\mathbf{x}_{-n}^{c\ T} \langle \mathbf{d}_{ki} \rangle + \langle w_{nki}^c \rangle \| \langle \mathbf{d}_k \rangle \|_2^2 \right) \\ &\quad + \langle \delta^c \rangle \left(\mathbf{q}_{-n}^{cT} \langle \mathbf{a}_{(k-1)L+i} \rangle + \langle w_{nki}^c \rangle \| \langle \mathbf{a}_{(k-1)L+i} \rangle \|_2^2 \right) \Big] \end{aligned} \tag{13}$$

Here, $\mathbf{x}_{-n}^c = \mathbf{x}_n^c - \mu^c - \sum_{k=1}^{K} \langle b_{nk}^c \rangle \langle \mathbf{d}_k \rangle * \langle \mathbf{w}_{nk}^c \rangle$ and $\mathbf{q}_{-n}^c = \mathbf{q}_n^c - \langle \mathbf{A} \rangle \langle \hat{\mathbf{w}}_n^c \rangle$ with $\langle \hat{\mathbf{w}}_n^c \rangle = \left[\langle b_{n1}^c \rangle \langle \mathbf{w}_{n1}^c \rangle^T, \langle b_{n2}^c \rangle \langle \mathbf{w}_{n2}^c \rangle^T, \cdots, \langle b_{nK}^c \rangle \langle \mathbf{w}_{nK}^c \rangle^T \right]^T$. According to the expressions in (13), the mean and variance of the posterior distribution for $\mathbf{w}_{nk}^c$ can be further derived. If we assume that $q(\mathbf{w}_{nk}^c) = \mathcal{N}(\mathbf{w}_{nk}^c | \varsigma_{nk}^c, diag(\Sigma_{nk}^c))$, the updating equations of ς_{nk}^c and Σ_{nk}^c are as follows:

$$\begin{aligned} \Sigma_{nk}^c &= \mathbf{1}_L \oslash \left(\langle \boldsymbol{\alpha}_{nk}^c \rangle + \langle \gamma^c \rangle \langle b_{nk}^c \rangle \left\langle \| \mathbf{d}_k \|_2^2 \right\rangle \mathbf{1}_L + \langle \delta^c \rangle \langle b_{nk}^c \rangle column\left(\left\langle \mathbf{A}_k^T \mathbf{A}_k \right\rangle \right) \right) \\ \varsigma_{nk}^c &= \Sigma_{nk}^c \langle b_{nk}^c \rangle \Big[\langle \gamma^c \rangle \left(\mathbf{x}_{-n}^{c\ T} * \langle \mathbf{d}'_k \rangle + \langle \mathbf{w}_{nk}^c \rangle \| \langle \mathbf{d}_k \rangle \|_2^2 \right) \\ &+ \langle \delta^c \rangle \left(column\left(\mathbf{Q}_{-n}^{c\ T} \langle \mathbf{A}_k \rangle \right) + \langle \mathbf{w}_{nk}^c \rangle column\left(\left\langle \mathbf{A}_k^T \mathbf{A}_k \right\rangle \right) \right) \Big] \end{aligned} \tag{14}$$

where $\mathbf{1}_L \in \mathbb{R}^L$ is a vector that the values of all elements are 1; $\alpha_{nk}^c = [\alpha_{nk1}^c, \cdots, \alpha_{nkL}^c]^T$, $\mathbf{Q}_{-n}^c = [\mathbf{q}_{-n}^c, \cdots, \mathbf{q}_{-n}^c] \in \mathbb{R}^{C \times L}$ and $\mathbf{d}'_k = [d_{kJ}, d_{kJ-1}, \cdots, d_{k1}]^{\mathrm{T}} \in \mathbb{R}^J$; the symbol $\odot$ is the element-wise division operator and $\odot$ is the element-wise product operator; $diag(\cdot)$ is the diagonalization operation and column$(\cdot)$ returns the main diagonal elements of the matrix in parentheses in the form of a column.

Section 3.2 indicates that the max-pooling operation is implemented on $\mathbf{w}_{nk}^c$ after its update is completed. In detail, the maximum element in each max-pooling area remains unchanged, and the others are set to 0.

3) For π_k^c: $q(\pi_k^c) = Beta\left(\pi_k^c \middle| \tilde{c}_k^c, \tilde{d}_k^c\right)$ is assumed, and then

$$\begin{aligned} \tilde{c}_k^c &= c_0 + \sum_{n=1}^{N} \langle b_{nk}^c \rangle \\ \tilde{d}_k^c &= d_0 + N - \sum_{n=1}^{N} \langle b_{nk}^c \rangle \end{aligned} \tag{15}$$

4) For b_{nk}^c: we assume that $q(b_{nk}^c) = Bernoulli\left(b_{nk}^c \middle| \frac{Ep_{nk}^c}{Ep_{nk}^c + Eb_{nk}^c} \right)$, and

$$\begin{aligned} Eb_{nk}^c &\propto \exp\langle \ln(1 - \pi_k^c) \rangle \\ Ep_{nk}^c &\propto \exp\Big\{ \langle \ln \pi_k^c \rangle + \langle \gamma^c \rangle (\langle \mathbf{w}_{nk}^c \rangle * \langle \mathbf{d}_k \rangle)^T \mathbf{x}_{n,-k}^c - \tfrac{1}{2} \langle \gamma^c \rangle \left\langle \| \mathbf{w}_{nk}^c * \mathbf{d}_k \|_2^2 \right\rangle \\ &\quad + \langle \delta^c \rangle \left(\sum_{i=1}^{L} \left\langle \mathbf{a}_{(k-1)L+i}^T \right\rangle \langle w_{nki}^c \rangle \right) \mathbf{q}_{-n}^c - \tfrac{1}{2} \langle \delta^c \rangle \left(\sum_{i=1}^{L} \left\langle \| \mathbf{a}_{(k-1)L+i} \|_2^2 \right\rangle \langle w_{nki}^{c\ 2} \rangle \right) \\ &\quad + \tfrac{1}{2} \langle \delta^c \rangle \sum_{i=1}^{L} \| \langle \mathbf{a}_{(k-1)L+i} \rangle \|_2^2 \langle w_{nki}^c \rangle^2 \Big\} \end{aligned} \tag{16}$$

where $\mathbf{x}_{n,-k}^c = \mathbf{x}_n^c - \sum_{m=1, m \neq k}^{K} \langle b_{nm}^c \rangle (\langle \mathbf{w}_{nm}^c \rangle * \langle \mathbf{d}_m \rangle)$.

5) For β_k: we have $q(\beta_{kj}) = Gamma\left(\beta_{kj}\middle|\widetilde{e},\widetilde{f}\right)$ with

$$\begin{aligned} \widetilde{e} &= e_0 + \tfrac{1}{2} \\ \widetilde{f} &= f_0 + \tfrac{1}{2}\left\langle d_{kj}^2 \right\rangle \end{aligned} \tag{17}$$

6) For $\mathbf{d}_k$: we have $q(\mathbf{d}_k) = \mathcal{N}(\mathbf{d}_k|\xi_k, diag(\Lambda_k))$ and the distribution parameters, i.e., ξ_k and Λ_k, should have a similar form to those of $\mathbf{w}_{nk}^c$ because the convolution operation satisfies the commutative law. In detail,

$$\begin{aligned} \Lambda_k &= \mathbf{1}_J\left(\langle\beta_k\rangle + \sum_{c=1}^{C}\sum_{n=1}^{N_c}\langle\gamma^c\rangle\left\langle \| \mathbf{w}_{nk}^c \|_2^2 \right\rangle \mathbf{1}_J\right) \\ \xi_k &= \Lambda_k \sum_{c=1}^{C}\sum_{n=1}^{N_c}\langle\gamma^c\rangle\left\langle b_{nk}^c\right\rangle\left(\left\langle \mathbf{x}_{-n}^c\right\rangle * \left\langle \mathbf{w'}_{nk}^c\right\rangle + \langle\mathbf{d}_k\rangle\left\langle \| \mathbf{w}_{nk}^c \|_2^2\right\rangle\right) \end{aligned} \tag{18}$$

where $\mathbf{1}_J \in \mathbb{R}^J$ is a vector that the values of all elements are 1; $\mathbf{w'}_{nk}^c \in \mathbb{R}^L$ represents the vertically flipped version of $\mathbf{w}_{nk}^c$, i.e., $\mathbf{w'}_{nk}^c = \left[w_{nkL}^c, w_{nkL-1}^c, \cdots, w_{nk1}^c\right]^{\mathrm{T}}$.

7) For λ_{ki}: we have $q(\lambda_{ki}) = Gamma\left(\lambda_{ki}\middle|\widetilde{g},\widetilde{h}\right)$ with

$$\begin{aligned} \widetilde{g}_{ki} &= g_0 + \tfrac{C}{2} \\ \widetilde{h}_{ki} &= h_0 + \tfrac{1}{2}\left\langle \| \mathbf{a}_{(k-1)L+i} \|_2^2 \right\rangle \end{aligned} \tag{19}$$

8) For $\mathbf{a}_{(k-1)L+i}$: we have $q\left(\mathbf{a}_{(k-1)L+i}\right) = \mathcal{N}\left(\mathbf{a}_{(k-1)L+i}\middle|\zeta_{(k-1)L+i}, \Delta_{(k-1)L+i}\right)$ with

$$\begin{aligned} \Delta_{(k-1)L+i} &= \left(\langle\lambda_{ki}\rangle\mathbf{I}_C + \sum_{c=1}^{C}\sum_{n=1}^{N_c}\langle\delta^c\rangle\left\langle b_{nk}^c\right\rangle\left\langle {w_{nki}^c}^2\right\rangle\mathbf{I}_C\right)^{-1} \\ \zeta_{(k-1)L+i} &= \Delta_{(k-1)L+i}\sum_{c=1}^{C}\sum_{n=1}^{N_c}\langle\delta^c\rangle\left\langle b_{nk}^c\right\rangle\left(\mathbf{q}_{-n}^c\left\langle w_{nki}^c\right\rangle + \tfrac{1}{2}\left\langle\mathbf{a}_{(k-1)L+i}\right\rangle\| \left\langle w_{nki}^c\right\rangle \|_2^2\right) \end{aligned} \tag{20}$$

9) For γ^c: we have $q(\gamma^c) = Gamma\left(\gamma^c\middle|\widetilde{p}^c,\widetilde{q}^c\right)$ and

$$\begin{aligned} \widetilde{p}^c &= p_0 + \tfrac{NP}{2} \\ \widetilde{q}^c &= q_0 + \tfrac{1}{2}\sum_{n=1}^{N_c}\left(\left\| \mathbf{x}_n^c - \sum_{k=1}^{K}\left\langle b_{nk}^c\right\rangle\left\langle\mathbf{w}_{nk}^c\right\rangle * \langle\mathbf{d}_k\rangle \right\|_2^2 + \sum_{k=1}^{K}\upsilon_k^c\right) \end{aligned} \tag{21}$$

where $\upsilon_k^c = \left\langle \| \mathbf{w}_{nk}^c * \mathbf{d}_k \|_2^2\right\rangle - \| \left\langle\mathbf{w}_{nk}^c\right\rangle * \langle\mathbf{d}_k\rangle \|_2^2$.

10) For δ^c: the form of $q(\delta^c)$ is similar to $q(\gamma^c)$.

$$\begin{aligned} q(\delta^c) &= Gamma\left(\delta^c\middle|\widetilde{u}^c, \widetilde{\nu}^c\right) \\ \widetilde{u}^c &= u_0 + \tfrac{NC}{2} \\ \widetilde{\nu}^c &= \nu_0 + \tfrac{1}{2}\sum_{n=1}^{N_c}\left(\| \mathbf{q}_n^c - \langle\mathbf{A}\rangle\left\langle\hat{\mathbf{w}}_n^c\right\rangle \|_2^2 + \sum_{k=1}^{K}\vartheta_k^c\right) \end{aligned} \tag{22}$$

with $\vartheta_k^c = \sum_{i=1}^{L}\left\langle b_{nk}^c\right\rangle\left(\left\langle \| \mathbf{a}_{(k-1)L+i} \|_2^2\right\rangle\left\langle {w_{nki}^c}^2\right\rangle - \| \left\langle\mathbf{a}_{(k-1)L+i}\right\rangle \|_2^2\left\langle w_{nki}^c\right\rangle^2\right)$.

In the model learning, the parameters are iteratively updated according to the preceding expressions until the lower bound $L(q(\Phi))$ converges. In this paper, we consider the convergence of $L(q(\Phi))$ when its variation rate between two iterations is below 10^{-5}. Algorithm 1 summarizes the process of the LCCFA-based model learning.

Algorithm 1. Parameter estimation of the LCCFA model

Input: $\{\mathbf{x}_n^c, \mathbf{q}_n^c\}_{n=1,c=1}^{N_c,\,C}$, θ

Output: The posteriors of model parameters, including $\{b_{nk}^c, \pi_k^c, \mathbf{d}_k, \mathbf{w}_{nk}^c, \mathbf{a}_{(k-1)L+i}, \alpha_{nk}^c, \beta_k, \lambda_{ki}, \gamma^c, \delta^c\}_{i=1,\,k=1,\,n=1,\,c=1}^{L,\;K,\;N_c,\;C}$

1: Initialize the posterior distributions of all estimated model parameters.
2: **while** not converged **do**
3:   **for** $k = 1$ to K **do**
4:     Update the posterior distribution of β_k with (17);
5:     Update the posterior distribution of $\mathbf{d}_k$ with (18);
6:     **for** $i = 1$ to L **do**
7:       Update the posterior distribution of λ_{ki} with (19);
8:       Update the posterior distribution of $\mathbf{a}_{(k-1)L+i}$ with (20);
9:     **end for**
10:     **for** $c = 1$ to C **do**
11:       Update the posterior distribution of π_k^c with (15);
12:       **for** $n = 1$ to N_c **do**
13:         Update the posterior distribution of b_{nk}^c with (16);
14:         Update the posterior distribution of α_{nk}^c with (12);
15:         Update the posterior distribution of $\mathbf{w}_{nk}^c$ with (13);
16:         Max-pooling operation for $\mathbf{w}_{nk}^c$;
17:       **end for**
18:     **end for**
19:   **end for**
20:   **for** $c = 1$ to C **do**
21:     Update the posterior distribution of γ^c with (21);
22:     Update the posterior distribution of δ^c with (22);
23:   **end for**
24: **end while**

According to definition, the time complexity of an algorithm is dependent on the highest order term of the times of operations in statements. For the statement 4 in Algorithm 1, based on the posterior expression of $\beta_k = [\beta_{k1}, \cdots, \beta_{kJ}]^T$, i.e., $q(\beta_{kj}) = Gamma\left(\beta_{kj}\middle|\widetilde{e}, \widetilde{f}\right)$ with $\widetilde{e} = e_0 + \frac{1}{2}$ and $\widetilde{f} = f_0 + \frac{1}{2}\left\langle d_{kj}^2\right\rangle$, the highest order term of operation times is TKJ if we posit that the lower bound is converged after T iterations. Analogically, the highest order terms of the number of operations for statements 5, 6, 8, 11, 13, 14, 15, 16, 21, and 22 are $TKP^2\sum_{c=1}^{C} N_c$, $TKPC$, $TK^2P^2C\sum_{c=1}^{C} N_c$, TKC, $TK^2PJ\sum_{c=1}^{C} N_c$, $TKP\sum_{c=1}^{C} N_c$, $TK^2P^2C\sum_{c=1}^{C} N_c$, $TKP\sum_{c=1}^{C} N_c$, $TKPJ\sum_{c=1}^{C} N_c$, and $TKPC\sum_{c=1}^{C} N_c$, respectively. Due to the maximum value of $TK^2P^2C\sum_{c=1}^{C} N_c$, the time complexity of Algorithm 1 is $O\left(TK^2P^2C\sum_{c=1}^{C} N_c\right)$.

4.2. Bayes classifier design and statistical classification with the LCCFA model

As discussed at the beginning of this Section, the essence of the Bayes classifier is the estimation of the class posterior PDF $p(c|\mathbf{x})$ $(c = 1, \cdots, C)$. According to Bayes theorem, we have $p(c|\mathbf{x}) \propto p(\mathbf{x}|c)p(c)$ with $p(\mathbf{x}|c)$ being the class-conditional likelihood function and $p(c)$ the class prior distribution. In addition, the non-informative prior is usually imposed on the $p(c)$, i.e., $p(1) = p(2) = \cdots = p(C)$. Therefore, the key to the classifier is turned to the estimation of the class-conditional likelihood function. Next, we provide an estimation method of the class-conditional likelihood function based on the LCCFA model.

Given class c, the class-conditional likelihood function of data $\mathbf{x}$ can be calculated via the following expression.

$$p(\mathbf{x}|c) = \iiint \mathcal{N}\left(\mathbf{x}\middle|\hat{\mathbf{D}}\hat{\mathbf{w}}^c + \mu^c, \frac{1}{\gamma^c}\mathbf{I}_P\right) q(\gamma^c)q(\hat{\mathbf{w}}^c)q\left(\hat{\mathbf{D}}\right)d\gamma^c d\hat{\mathbf{w}}^c d\hat{\mathbf{D}} \tag{23}$$

where $\hat{\mathbf{D}} = [\mathbf{d}_{11}, \cdots, \mathbf{d}_{1L}, \cdots, \mathbf{d}_{K1}, \cdots, \mathbf{d}_{KL}] \in \mathbb{R}^{P\times KL}$ with $\mathbf{d}_{ki}$ $(k = 1, \cdots, K, i = 1, \cdots, L)$ representing the circularly shifted version of $\mathbf{d}_k$ by $i - 1$. $q\left(\hat{\mathbf{D}}\right)$ and $q(\gamma^c)$ are the estimated posterior distributions for $\hat{\mathbf{D}}$ and γ^c that can be obtained after learning the LCCFA model. $\hat{\mathbf{w}}^c$ is the statistical weight of class c, the posterior of which can be simplified as

$$
\begin{aligned}
& q(\hat{\mathbf{w}}^c) = \mathcal{N}(\hat{\mathbf{w}}^c | \eta^c, \Omega^c) \\
& \eta^c = \tfrac{1}{N_c} \sum_{n=1}^{N_c} \langle \hat{\mathbf{w}}_n^c \rangle \\
& \Omega^c = \tfrac{1}{N_c} \sum_{n=1}^{N} \left(\langle \hat{\mathbf{w}}_n^c \rangle - \eta^c \right) \left(\langle \hat{\mathbf{w}}_n^c \rangle - \eta^c \right)^T
\end{aligned} \tag{24}
$$

Here, the posterior expectation $\langle \hat{\mathbf{w}}_n^c \rangle$ is the concatenation of all weight vectors corresponding to $\mathbf{x}_n^c$, i.e., $\hat{\mathbf{w}}_n^c = \left[b_{n1}^c \mathbf{w}_{n1}^{c\mathrm{T}},\ b_{n2}^c \mathbf{w}_{n2}^{c\mathrm{T}}, \cdots, b_{nK}^c \mathbf{w}_{nK}^{c\mathrm{T}} \right]^{\mathrm{T}}$. Actually, it is hard to calculate the value of the triple integral in (23). Hence, we adopt the following expression as an alternative approach to approximate (23):

$$
p(\mathbf{x}|c) = \int \mathcal{N}\left(\mathbf{x} \middle| \left\langle \hat{\mathbf{D}} \right\rangle \hat{\mathbf{w}}^c + \mu^c, \langle \gamma^c \rangle^{-1} \mathbf{I}_P \right) q(\hat{\mathbf{w}}^c) d\hat{\mathbf{w}}^c \tag{25}
$$

with $\left\langle \hat{\mathbf{D}} \right\rangle$ and $\langle \gamma^c \rangle$ being the posterior expectations of $\hat{\mathbf{D}}$ and γ^c, respectively. A similar estimation method is used in [4].

According to (25), the class-conditional likelihood function of $\mathbf{x}$ given class c can be calculated as

$$
\begin{aligned}
& p(\mathbf{x}|c) = \mathcal{N}(\mathbf{x} | \chi^c, \Gamma^c) \\
& \chi^c = \mu^c + \left\langle \hat{\mathbf{D}} \right\rangle \eta^c \\
& \Gamma^c = \left\langle \hat{\mathbf{D}} \right\rangle \Omega^c \left\langle \hat{\mathbf{D}}^c \right\rangle^T + \langle \gamma^c \rangle^{-1} \mathbf{I}_P
\end{aligned} \tag{26}
$$

Furthermore, for a test sample $\mathbf{x}*$, the Bayes classifier calculates its class-conditional likelihood under all classes with (24) and (26), and then assign it to the class with the maximum class-conditional likelihood.

5. Experimental results

In this section, two types of experiments are conducted to assess the performance of the proposed LCCFA model. The first experiment is the data classification task on several benchmark datasets. In the second experiment, we apply the LCCFA model to radar high-resolution range profile (HRRP) target recognition, where sufficient training data are often unavailable, and Section 5.2 will give a detailed description. We demonstrate the superiority of our method by comparing with the following statistical approaches: subspace approximation model based on PCA (referred to as PCA), PPCA, STL-FA, MTL-FA, convolutional FA model without label constraint (denoted as CFA), and our LCCFA. In the STL-FA model, the number of dictionary atoms is determined by the following approach: firstly, a candidate range $\mathcal{M}$ for the number of latent factors V is pre-

Table 2
Description of the four benchmark datasets and the setting of training sample number.

Dataset	Class number	Data dimension	Data size	Settings of r and their corresponding N'	
				r	N'
Heart	2	13	270	0.48	3
				1.08	7
				1.54	10
				2.00	13
				2.46	16
				3.08	20
Ringnorm	2	20	7400	0.50	5
				1.00	10
				1.50	15
				2.00	20
				2.50	25
				3.00	30
Twonorm	2	20	7400	0.50	5
				1.00	10
				1.50	15
				2.00	20
				2.50	25
				3.00	30
Waveform	3	21	5000	0.57	4
				0.71	5
				1.29	9
				1.57	11
				2.29	16
				3.00	21

specified; secondly, for each $V \in \mathcal{M}$, a FA model is independently trained for each class; lastly, the optimal value of V, i.e., V_{opt}, is defined as the dictionary atom number that corresponds to the highest recognition accuracy.

5.1. Classification on benchmark data

5.1.1. Introduction of benchmark datasets

In this part, four benchmark datasets from the UCI Machine Learning Repository, including Heart [6,47], Ringnorm [47], Twonorm [6,47] and Waveform [6,47], are used to validate LCCFA model's effectiveness. For each dataset, we randomly divide it into two equal-sized parts, one of which are selected to training the model, i.e., training set, and another part is used as the test set. In the experiment, we repeat the data division 20 times to deal with the uncertainty caused by random partitions, and the results of 20 partitions are averaged.

After data division for each dataset, we test the performance of our LCCFA model and other related methods in the case of different training data sizes. In this paper, we control the number of training samples by adjusting the ratio of the total training sample number to data dimension, i.e., r. Here, $r = \frac{CN'}{P}$ with P, C and N' denoting the data dimension, the number of classes for each dataset, and the training sample number of each class, respectively. For each dataset, the detailed introduction and the settings of r with their corresponding N' are presented in Table 2.

5.1.2. Classification performance and result analysis

Fig. 3 presents the test accuracies of different methods for each dataset versus the ratio r, from which we see that the proposed LCCFA has greater performance advantages in classification than the other methods. For PCA and PPCA models,

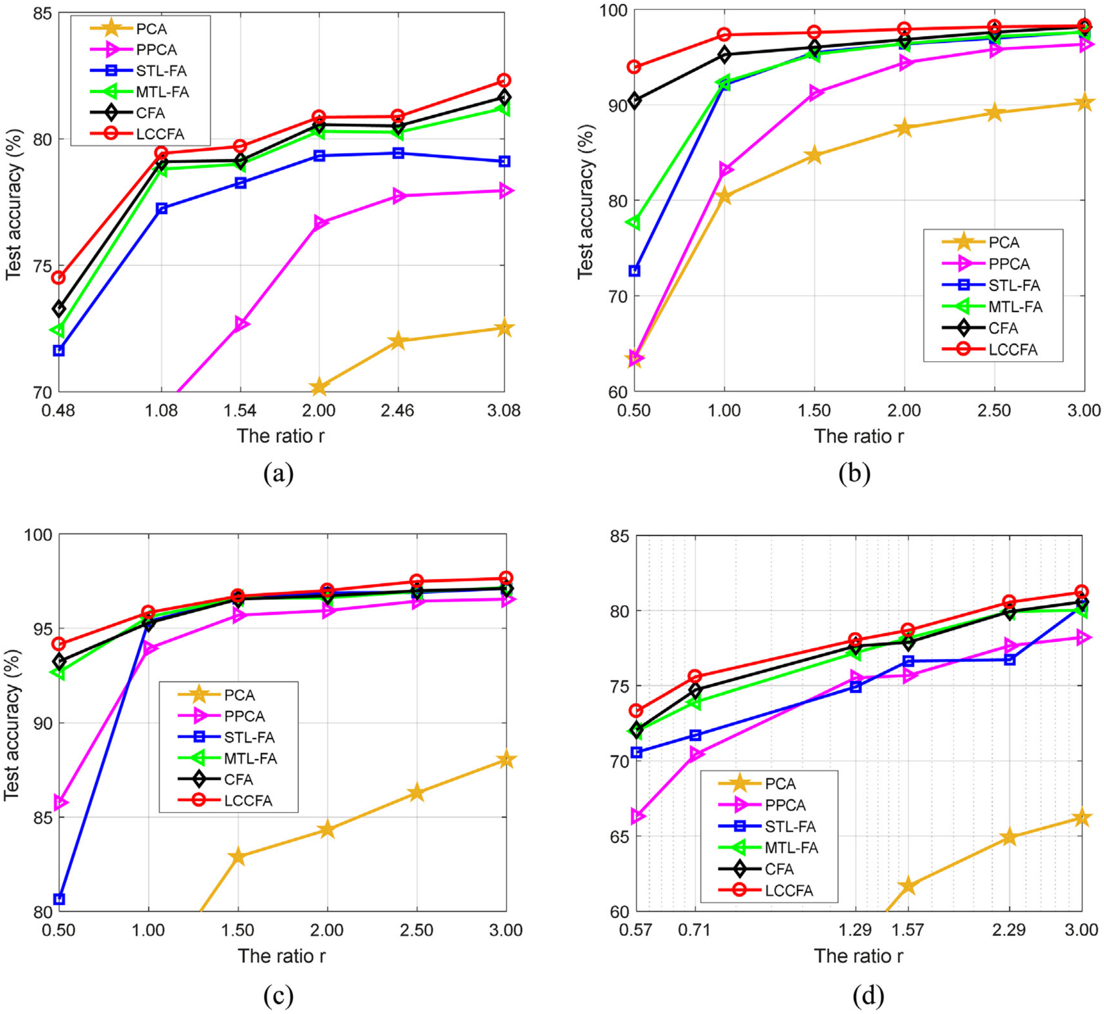

Fig. 3. Variations of the test accuracy with r on different datasets. (a) Heart; (b) Ringnorm; (c) Twonorm; (d) Waveform.

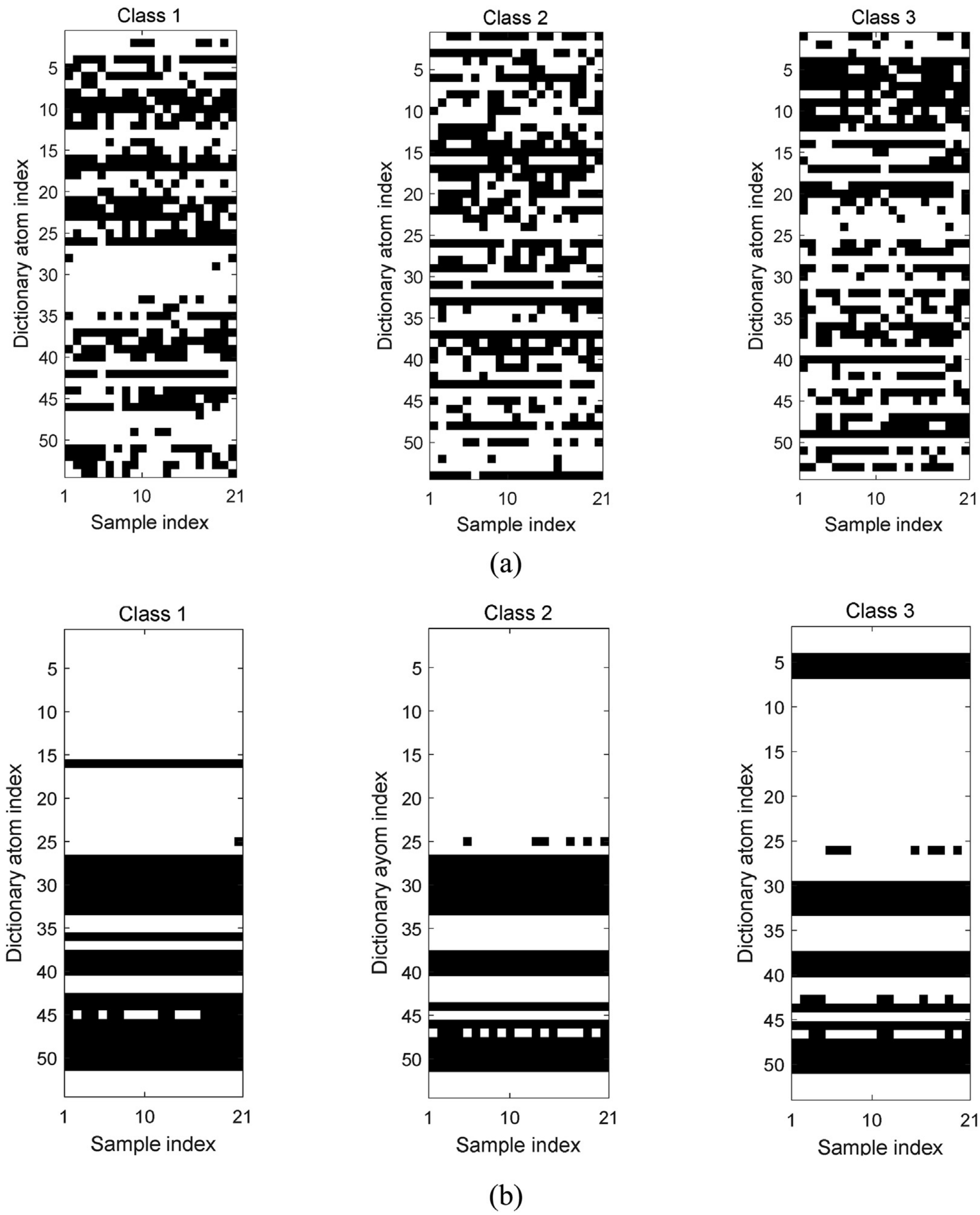

Fig. 4. Dictionary atom usage for 21 training samples per class, generated via the runs of the MTL-FA and LCCFA models on Waveform dataset, i.e., $\left\langle z^{c}_{n,v}\right\rangle$ and $\langle b^{c}_{nk}\rangle$ with $c = 1, 2, 3$ and $n = 1, \cdots, 21$. (a) MTL-FA model; (b) LCCFA model.

they both need to estimate a sample covariance matrix of size $P \times P$. Rather than giving a direct estimation of sample covariance matrix, the FA-based models and our convolutional models only need to learn the small-scale dictionary and noise power, which reduces the number of model parameters to be estimated. Therefore, the PCA model and PPCA model have the lower classification accuracies than the other four models when sufficient training data are unavailable, as shown in Fig. 3. Moreover, the MTL-based statistical models, i.e., MTL-FA, CFA, and LCCFA, outperform the STL-FA model owing to their fewer model parameters to be estimated by sharing a dictionary among all training samples. Meanwhile, the classification accuracy rates of the CFA model are higher than those of the MTL-FA model, because the introduction of convolution operation facilitates the CFA model to extract observations' basic structures, which offers the potential to describe the observed data with fewer dictionary atoms and accurately estimate parameters by using a few training data. Compared with the CFA model, the LCCFA model utilizes the class labels of observations to restrict the learning of the model parameters. The introduction of additional information enables the enhancement of the separability of statistical models from different classes.

Therefore, the LCCFA model has better classification accuracy than the CFA model. In summary, the proposed LCCFA model is more suitable for the small sample classification problem.

In the MTL-based statistical models, the proper number of dictionary atoms can be inferred via the BP prior. Fig. 4 presents the model selection results of the MTL-FA and LCCFA models on Waveform dataset with $r = 3.00$ and the total numbers of dictionary atoms of the two model being set to 54, i.e., $\left\langle z^c_{n,v} \right\rangle$ and $\langle b^c_{nk} \rangle$ with $n = 1, \cdots, 21$, $v = 1, \cdots, 54$, $k = 1, \cdots, 54$ and $c = 1,\ 2,\ 3$. In Fig. 4, if the value of $\left\langle z^c_{n,v} \right\rangle$ or $\langle b^c_{nk} \rangle$ is larger than 0.5, we show it in black to represent that the vth or kth dictionary atom contributes to the description of $\mathbf{x}^c_n$; otherwise, we show it in white. Fig. 4 exhibits that only a portion of the dictionary atoms are selected for each sample, which demonstrates that the initial number of dictionary atoms ($V = K = 54$) is sufficient enough to represent the observations from all classes. Meanwhile, the dictionary atoms used to describe the observations are different for different classes. Therefore, the binary selection vector can automatically select the suitable dictionary atoms for the corresponding observed data. Moreover, although the selected dictionary atoms for the samples from each class are not exactly the same, some different samples may select the same dictionary atoms, thereby confirming the rationality of sharing dictionary atoms among all training samples. In addition, due to the introduction of convolution operation, the dictionary atoms in our LCCFA model can reflect observations' basic information. The dictionary atoms are often highly shared when their excavated information is basic. Therefore, the dictionary atoms selected for a class in our LCCFA model are more likely to be selected for other classes in comparison with those of the MTL-FA model, which can be justified in Fig. 4. This property of the proposed method is propitious to represent data with the fewer dictionary atoms. The probability distributions on the required dictionary atom number for the MTL-FA and LCCFA models in the same experiment are presented in Fig. 5. Note from Fig. 5 that the histogram of the MTL-FA model peaks at 24 dictionary atoms, while 17 atoms are required for the majority of the samples in the LCCFA model, and even the maximum number of the required atoms is only 22. The smaller the number of dictionary atoms is, the lower the model complexity is. Therefore, our LCCFA model can obtain better classification performance than the MTL-FA model under a limited training sample condition.

To validate the separability strength of the LCCFA, we plot the average KL distances of the inter-class statistical models under different r values, via the MTL-FA, CFA and LCCFA models in Fig. 6. For the two statistical models, the similarity of the probability distributions that they describe, i.e., $p_1(\mathbf{x})$ and $p_2(\mathbf{x})$, can be measured by KL distance [38]. In detail, the KL distance between $p_1(\mathbf{x})$ and $p_2(\mathbf{x})$ is defined as

$$KL(p_1(\mathbf{x}) \| \, p_2(\mathbf{x})) = \int (p_1(\mathbf{x}) - p_2(\mathbf{x})) \ln \frac{p_1(\mathbf{x})}{p_2(\mathbf{x})} d\mathbf{x} \tag{27}$$

Obviously, the more similar the two models are, the closer the two probability distributions are, and the smaller the KL distance is. Specially, the KL distance is equal to 0 when $p_1(\mathbf{x}) = p_2(\mathbf{x})$. In this paper, the KL distance of any two statistical models can be calculated according to (27). The results in Fig. 6 show that our LCCFA model has the larger inter-class KL distance than the MTL-FA model and CFA model, demonstrating the effectiveness of label constraint on the enhancement of separability.

As presented in Section 4.1, the time complexity of the LCCFA-based model learning can be estimated. Analogically, the time complexities during the training process for the CFA, MTL-FA, STL-FA, PCA, PPCA models can also be calculated. For an intuitive comparison, the estimation of the time complexity from different methods are shown in Table 3, where $T_{\text{MTL - FA}}$, T_{CFA}, and T_{LCCFA} are the numbers of iterations when the lower bounds converge for the MTL-FA, CFA, and LCCFA models, respectively. Since an FA model is constructed for each class in the STL-FA model, we use $T^c_{\text{STL - FA}}$ to represent the required iteration number for the model learning of the cth class. Moreover, $K_{\text{MTL - FA}}$, K_{CFA}, and K_{LCCFA} are the numbers of shared dictionary atoms for the MTL-FA, CFA, and LCCFA models, respectively.

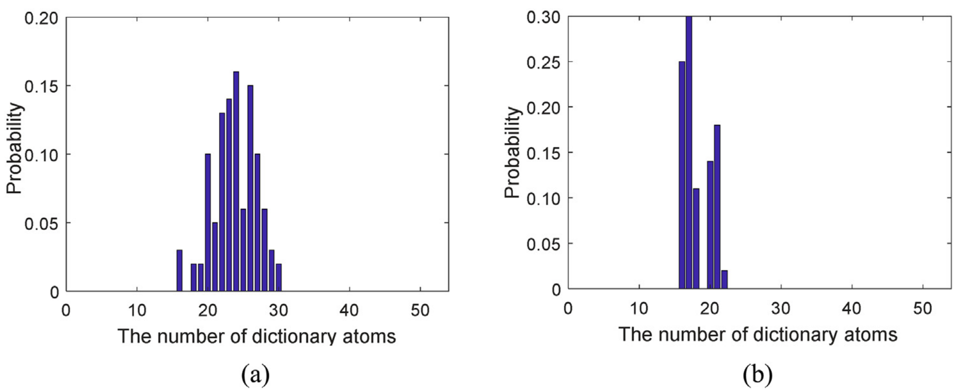

Fig. 5. Probability distributions on the required dictionary atom number for the MTL-FA model and the LCCFA model on Waveform dataset. (a) MTL-FA model; (b) LCCFA model.

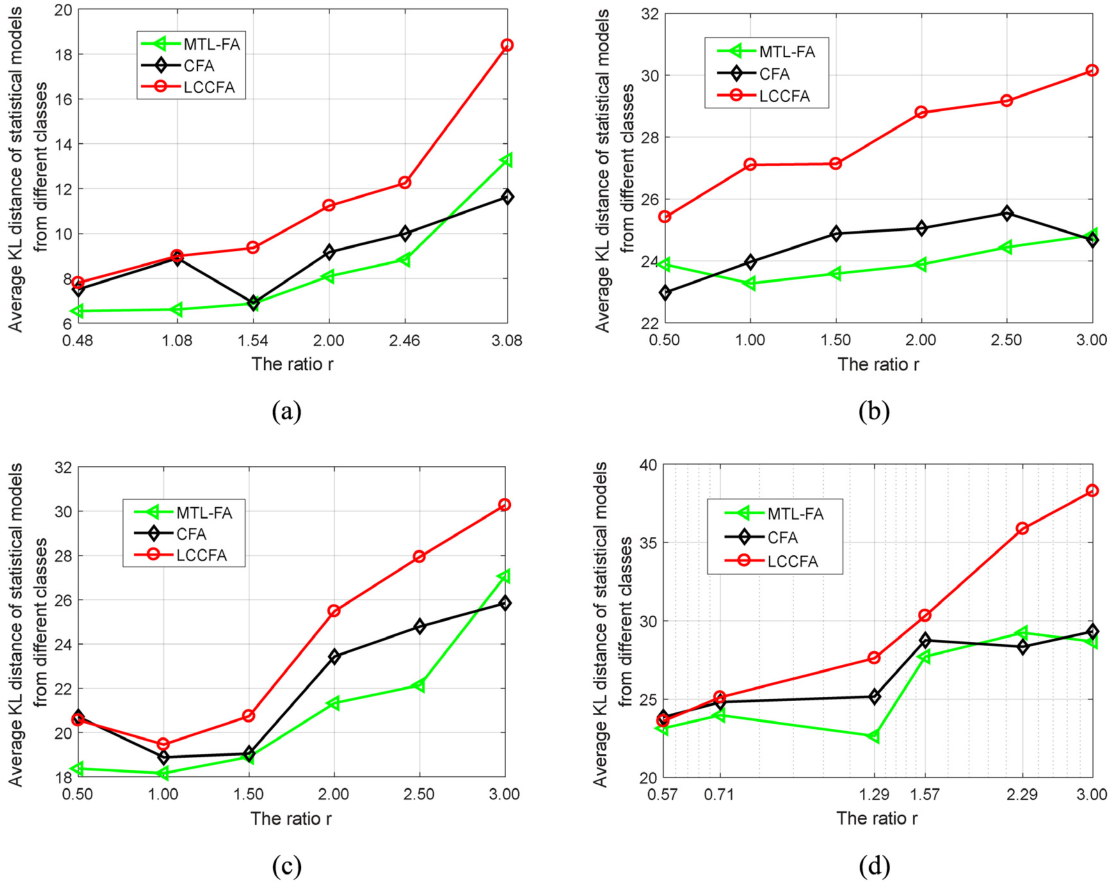

Fig. 6. Average KL distances of inter-class statistical models under different r values, via the MTL-FA, CFA, and LCCFA models. (a) Heart; (b) Ringnorm; (c) Twonorm; (d) Waveform.

Table 3
Time complexities during the training stage for the different models.

Method	Time complexity
PCA	$O\left(P^2\sum_{c=1}^{C}N_c\right)$
PPCA	$O\left(P^2\sum_{c=1}^{C}N_c\right)$
STL-FA	$O\left(P^2\sum_{c=1}^{C}T_{\text{STL - FA}}^{c}N_c\right)$
MTL-FA	$O\left(T_{\text{MTL - FA}}K_{\text{MTL - FA}}^{3}\sum_{c=1}^{C}N_c\right)$
CFA	$O\left(\max\left\{T_{\text{CFA}}K_{\text{CFA}}^{2}PJ,\ \ T_{\text{CFA}}K_{\text{CFA}}P^2\sum_{c=1}^{C}N_c\right\}\right)$
LCCFA	$O\left(T_{\text{LCCFA}}K_{\text{LCCFA}}^{2}P^2C\sum_{c=1}^{C}N_c\right)$

Table 4
Training time (s) of the different methods.

Training sample number per class	4	5	9	11	16	21
PCA	3.0×10^{-4}	5.0×10^{-4}	1.5×10^{-3}	2.0×10^{-3}	4.0×10^{-3}	5.5×10^{-3}
PPCA	8.0×10^{-4}	1.0×10^{-3}	2.2×10^{-3}	3.4×10^{-3}	4.9×10^{-3}	7.1×10^{-3}
STL-FA	0.7	1.0	2.3	2.9	4.3	5.9
MTL-FA	15.6	20.8	40.3	70.9	95.2	120.8
CFA	9.2	10.3	12.3	24.3	39.3	55.7
LCCFA	12.3	15.1	34.1	54.3	80.1	103.6

Table 3 shows that PCA and PPCA has the lowest time complexities because these two models only need to estimate the sample covariance matrix with N_c $(c = 1, \cdots, C)$ P-dimensional samples for each class, whose solution is non-iterative. For the STL-FA model, its time complexity is higher than those of the PCA and PPCA models due to the iterative solution of VB algorithm but lower than those of the MTL-based methods, i.e., MTL-FA, CFA, and LCCFA. The LCCFA model gives a description to observations, and at the same time it also describes the sample labels, which increases the amount of calculation. Therefore, the LCCFA model has the larger time complexity than the CFA model (usually, $T_{\text{CFA}} \approx T_{\text{LCCFA}}$), as shown in Table 3. However, it is difficult to compare the time complexities of the LCCFA and MTL-FA models due to their different values of iteration number and shared dictionary atoms for diverse datasets. In Table 4, we take the Waveform dataset as an example and give the comparison of the training time in the case of different sample numbers for above six methods, from which we note that the MTL-based methods (MTL-FA, CFA, and LCCFA) take much more time to perform model learning than the STL-based models (PCA, PPCA, and STL-FA). The PCA and PPCA models have the lowest demand for the training time. Moreover, the LCCFA model requires the higher run time cost than the CFA model, however, it spends less time than the MTL-FA model on Waveform dataset.

In summary, the proposed method obtains better classification performance at the expense of higher time complexity during the training process. Nevertheless, in the actual classification task, the model learning is performed off-line, and only the class-conditional probability of a test sample is required to be calculated via the Bayes classifier, which depends on the class mean and class covariance matrix estimated by using the learned model parameters in the training stage. Due to the same sizes of class mean and class covariance (only related to the sample dimension) for all methods, they have equal time complexities in the calculation of class-conditional likelihood, such as about 0.08s on Waveform dataset for all methods.

5.2. Radar HRRP target recognition

5.2.1. Data description

Radar target can be described with the scattering center model [10,11]. This model considers the target echo along with the direction of radar light-of-sight (LOS) as the coherent summations of complex time returns from target scatterers in each range resolution cell, whose amplitude is defined as the radar high-resolution range profile (HRRP). An example of HRRP sample is presented in Fig. 7. Due to the reflection of target structures, such as shape and size, the topic of HRRP-based radar target recognition is widely attractive [9–13,35]. However, a key issue for radar HRRP target recognition is that sufficient training samples usually cannot be guaranteed because of the limitation of radar sampling rate and the great change of flight attitude for the target with high velocity. Therefore, the accurate recognition of targets with small training HRRP sample supply has become a hotpot, and the proposal of our LCCFA model is of practical significance. In the following, we will give a detailed performance analysis of the LCCFA model in the HRRP-based radar target recognition.

Some sensitivity problems, such as target-aspect, time-shift, and amplitude-scale sensitivities, should be addressed before HRRP is utilized to train the LCCFA model. To deal with the target-aspect sensitivity, the aspect-frame partition method [13] is adopted in this paper. In detail, the whole HRRP sample set from each target is divided into several parts, and the target-aspect variation of HRRPs from each part is smaller than that corresponding to the scatterers' motion through range cells (MTRC). We refer each HRRP part as an aspect-frame. Suppose that there are C targets and the cth $(c = 1, \cdots, C)$ target has M_c aspect-frames, the STL-FA model independently learns a FA model for each aspect-frame HRRP data from each target. The MTL-FA or our LCCFA model refers to the joint learning of the models for all aspect-frames of all targets with the

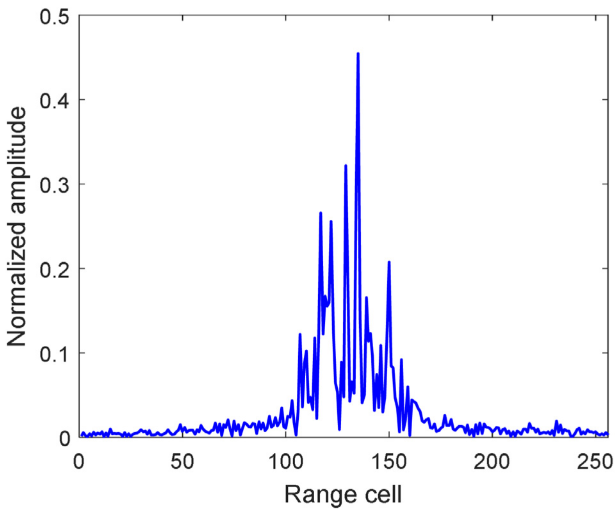

Fig. 7. Example of an HRRP waveform.

Table 5
Parameters of three airplanes and radar in the experiment.

Radar parameters	Center frequency Bandwidth	5520*MHz* 400*MHz*	
Airplanes	Length (m)	Width (m)	Height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

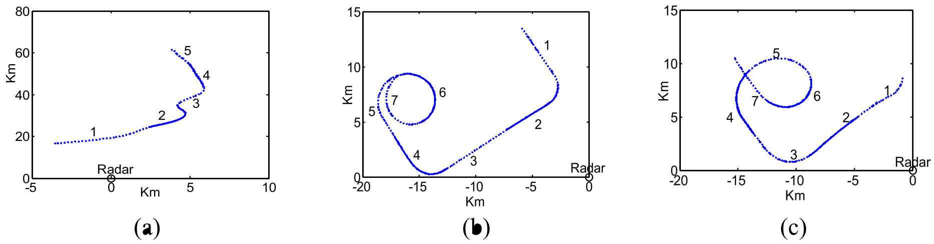

Fig. 8. Two-dimensional track charts of three radar targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

dictionary atoms being shared among all HRRP data. In the classification stage, the class-conditional likelihood of a test sample $\mathbf{x}^*$ given target c is $p(\mathbf{x}^*|c) = \max_m p(\mathbf{x}^*|m, c)$, where $p(\mathbf{x}^*|m, c)$ denotes the frame-conditional likelihood of $\mathbf{x}_*$ given the aspect-frame m of target c and can be estimated by referring to (24) and (26); m denotes the aspect-frame index with $m = 1, \cdots, M_c$. Finally, the class corresponding to the maximum class-conditional likelihood is considered as the target class that $\mathbf{x}^*$ belongs to. In addition, the time-shift sensitivity of HRRP can be overcome via some motion compensation technologies. In our paper, sliding correlation processing [13] is firstly used for each aspect-frame to align the intra-frame HRRPs. Then, we circularly shift the aligned HRRP samples by the same number of range cells to make the average profile's barycenter of each frame at the centre. To handle the amplitude-scale sensitivity of HRRP, each HRRP is divided by its amplitude, i.e., L_2 normalization [13,35].

In the experiment, the measured HRRP data recorded by radar are from three real airplanes: Yark-42, Cessna Citation S/II, and An-26. Table 5 presents the airplanes' sizes and radar measurement parameters. The two-dimensional track charts of three targets are shown in Fig. 8, from which the target-aspect of each target can be obtained based on the angle between the directions of the radar LOS and target's velocity. Note from Fig. 8 that each target track is split into several segments. In each segment, there are 25,600 HRRP samples with each sample of size 256×1, i.e., $P = 256$. We select the training and test datasets according to the following criterions: *a)* the target-aspect of the training samples should cover almost all those of the test samples; *b)* training and test samples' elevation angles should be different. Based on the requirements above, we consider the 2th and 5th segments from Yark-42, the 6th and 7th segments from Cessna Citation S/II, the 5th and 6th segments from An-26 as the training dataset, and other data segments as the test dataset. As mentioned before, there are several

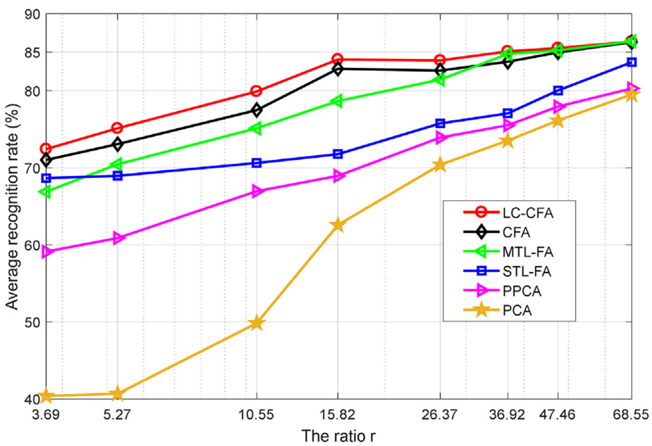

Fig. 9. Recognition performance under different values of r, via the PCA, PPCA, STL-FA, MTL-FA, CFA, and LCCFA models.

aspect-frames for each target. In our experiment, Yark-42, Cessna Citation S/II 50 frames and An-26 contain 35, 50, 50 frames, respectively (i.e., $C = 3$, $M_1 = 35$, $M_2 = M_3 = 50$), and each frame has 1024 HRRP samples ($N_c = 1024$, $c = 1, 2, 3$).

The performance of our method under different training data sizes is tested. Similar to Section 5.1, we also control the training sample number by adjusting the ratio of the number of the total training samples to sample dimension, i.e., $r = \sum_{c=1}^{C} M_c \cdot N'/P$, where N' denotes the number of training samples in each frame. In this subsection, we consider eight training datasets of different sizes to conduct experiments: $r = 3.69$, $r = 5.27$, $r = 10.55$, $r = 15.82$, $r = 26.37$, $r = 36.92$, $r = 47.46$, and $r = 68.55$, each corresponding to $N' = 7$, $N' = 10$, $N' = 20$, $N' = 30$, $N' = 50$, $N' = 70$, $N' = 90$, and $N' = 130$, respectively. In addition, the 17-dimensional dictionary atom, i.e., $J = 17$, is used for our LCCFA model.

5.2.2. Target recognition performance and training result analysis

Fig. 9 shows a comparison of the recognition accuracy with different training data sizes (i.e., different r values) for the LCCFA, CFA, MTL-FA, STL-FA, PPCA, and PCA models. It is observed that the increase of training samples can improve the recognition rate for each model, and moreover, the LCCFA model exhibits the superior recognition performance compared with other models, especially when r varies from 3.69 to 26.37. If we take 80% as a reference threshold, the recognition rate of our LCCFA model can satisfy the aforementioned requirement only by using 20 ($r = 10.55$) training samples per frame. Nevertheless, the CFA, MTL-FA, STL-FA, PPCA, and PCA models need about 30 ($r = 15.82$), 50 ($r = 26.37$), 90 ($r = 47.46$), 130 ($r = 68.55$), and 130 ($r = 68.55$) training samples per frame, respectively. Assuming that the minimum acceptable recognition rate is 70% , both the LCCFA and CFA models can efficiently work under the condition of 7 training samples per frame ($r = 3.69$), but the recognition rate of the CFA model is approximately 1.5% lower than that of the LCCFA model. The MTL-FA, STL-FA, PPCA, and PCA models, however, require approximately 10, 20, 30, and 50 training samples per frame, respectively. Consequently, the proposed LCCFA model is desirable for solving the problem with small-sized training data.

To further verify that the dictionary atoms learned by our LCCFA model can extract more basic structures from data, we train the MTL-FA and LCCFA models using 1350 HRRP samples (i.e., $r = 5.27$), and show the inferred dictionary atoms in Fig. 10(a) and (b), respectively. It should be pointed out that 48 and 15 dictionary atoms are selected to represent the observations in the MTL-FA and LCCFA models, respectively. For convenience, we only plot six dictionary atoms with high sharing rate for each model. The sharing rate of the tth dictionary atom is defined as $\frac{n_t}{N}$, where n_t denotes the number of samples selecting the tth dictionary atom and N denotes the total training sample number. In addition, 128 elements close to the support for each dictionary atom learned by the MTL-FA model are shown in Fig. 10(a) for clarity, and the sharing rates of the dictionary atoms decrease from left to right and top to bottom.

Note from Fig. 10 that the learned dictionary atoms in our LCCFA model are composed of lines and a few spikes, which are simple primitives of HRRP (the HRRP sample consists of some spikes as shown in Fig. 7). By contrast, the shapes reflected by the dictionary atoms in the MTL-FA model are relatively complex, thereby leading to the poor sharing among observations. Actually, the sharing rates of the atoms in Fig. 10(a) and (b) are 50.81%, 49.63%, 47.85%, 46.07%, 45.78%, 44.44%, and 100%, 99.33%, 99.26%, 97.70%, 97.02%, 96.58%, respectively. The lower the sharing is, the more atoms the model needs. In Fig. 11, we present the numbers of the dictionary atoms learned by the MTL-FA and LCCFA models with different r values, from which we see that the dictionary atom number required to be learned by the MTL-FA model is much larger than that by the LCCFA model.

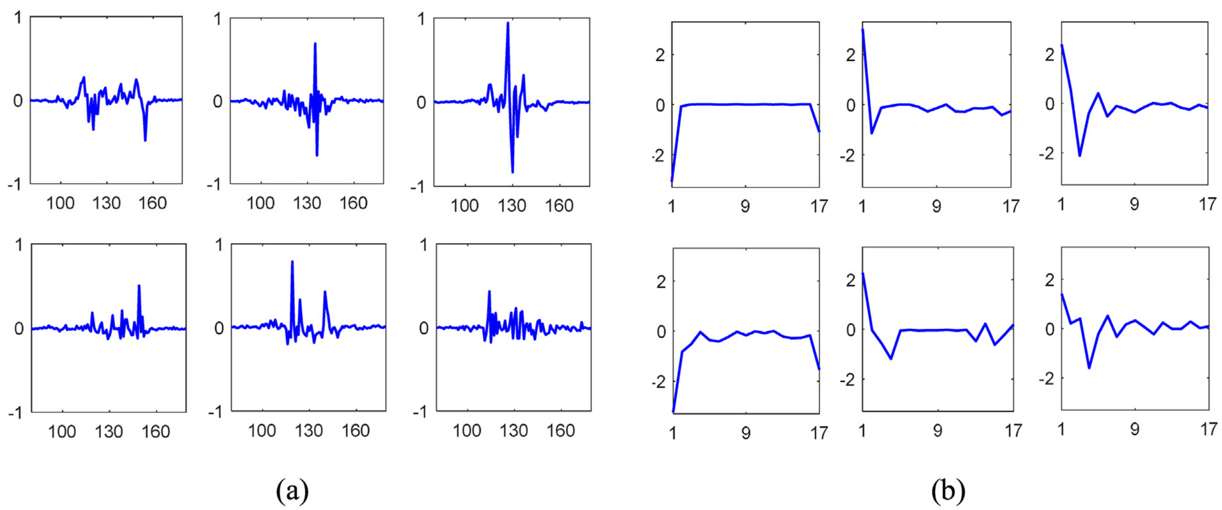

Fig. 10. The learned dictionary atoms with high sharing rates, via the MTL-FA and LCCFA models. (a) MTL-FA model; (b) LCCFA model.

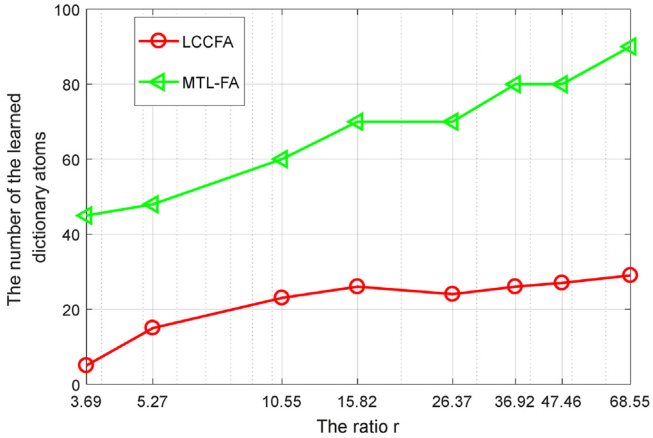

Fig. 11. The number of dictionary atoms required to be learned by the MTL-FA and LCCFA models, versus different r values.

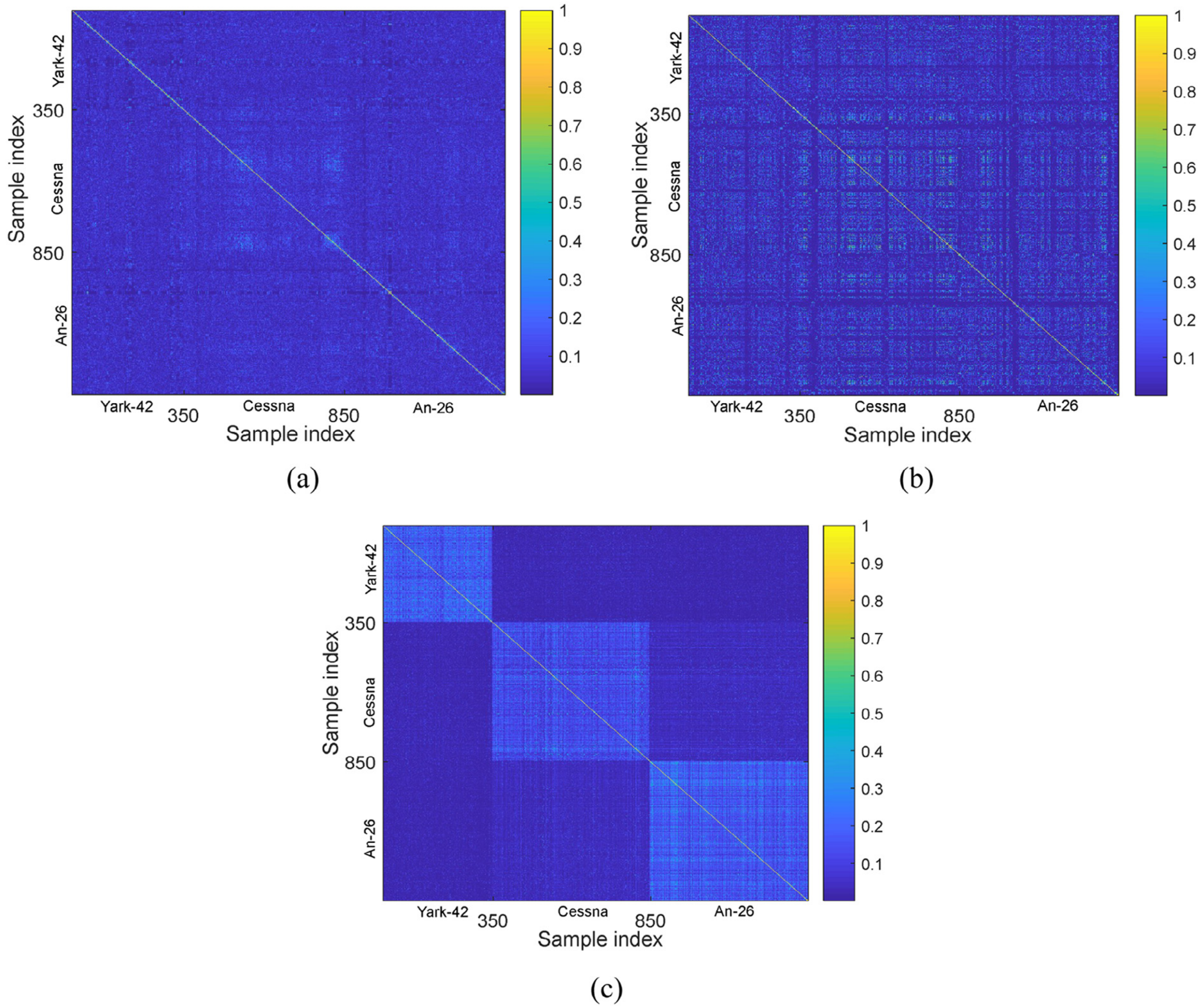

Fig. 12. Graph representations of similarity matrices via different methods with $r = 5.27$. (a) MTL-FA model; (b) CFA model; (c) LCCFA model.

Table 6
ACCs of MTL-FA, CFA, and LCCFA models with $r = 5.27$.

ACC	Methods		
	MTL-FA	CFA	LCCFA
$\mathrm{ACC}_{\mathrm{intra}}$	0.0537	0.0635	0.1361
$\mathrm{ACC}_{\mathrm{inter}}$	0.0535	0.0604	0.0361

Table 7
R_{KL} of the MTL-FA, CFA, and LCCFA models under different r values.

Values of r	3.69	5.27	10.55	15.82	26.37	39.62	47.46	68.55
MTL-FA	0.81	0.79	0.78	0.68	0.69	0.67	0.66	0.64
CFA	0.70	0.66	0.65	0.63	0.62	0.61	0.62	0.59
LCCFA	0.58	0.56	0.54	0.52	0.49	0.47	0.46	0.45

In LCCFA-based HRRP statistical modeling, the weight vectors of the samples from all aspect-frames of each target are mapped to their corresponding class labels. This technique strengthens the distinctions of the weight vectors from the different classes and is propitious to the improvement of inter-class separability of the statistical models. In the following, we will make an analysis on the effect of the label constraint.

We train the MTL-FA, CFA, and LCCFA models by using 1350 HRRP samples, i.e., $r = 5.27$, and present the similarity matrix graphs of weight vectors in Fig. 12. The element at the xth row and yth column s_{xy} $(x = 1, \cdots, 1350$ and $y = 1, \cdots, 1350)$ in the similarity matrix represents the correlation coefficient of two weight vectors and is defined as

$$s_{xy} = \frac{\tilde{\mathbf{w}}_x^T \tilde{\mathbf{w}}_y}{\| \tilde{\mathbf{w}}_x \|_2 \| \tilde{\mathbf{w}}_y \|_2} \tag{28}$$

where $\tilde{\mathbf{w}}_x$ denotes the weight vector of the xth HRRP sample and similar meaning for $\tilde{\mathbf{w}}_y$. In Fig. 12, the first 350 (i.e., $x = 1, \cdots, 350$ and $y = 1, \cdots, 350$) weight vectors are from Yark-42, the latter 500 (i.e., $x = 851, \cdots, 1350$ and $y = 851, \cdots, 1350$) weight vectors are from An-26, and the rest of weight vectors are from Cessna Citation S/II. The larger the value of s_{xy} is, the more relevant the two weight vectors are. Compared with the MTL-FA and CFA models, the similarity matrix of the LCCFA model presents significantly partitioned phenomena as shown in Fig. 12, that is, the weight vectors from the same class are more similar and those from different classes are more discriminating. Table 6 quantitatively shows the average correlation coefficients (ACCs) of the intra-class weight vectors and inter-class weight vectors. For convenience, we abbreviate intra-class and inter-class ACCs to $\mathrm{ACC}_{\mathrm{intra}}$ and AAC to $\mathrm{ACC}_{\mathrm{inter}}$, respectively. The results shown in Table 6 further demonstrate the effectiveness of label constraint on the learning of the model parameters.

The enhancement of the differences among weight vectors from different classes helps strengthen the model's discriminative ability. Similar to Section 5.1.2, we exploit the KL distance to measure the similarity of the two models. Since the HRRP dataset from a target is divided into several aspect-frames and the statistical characteristic of each frame can be learned via the corresponding model, we use average KL ratio, represented as R_{KL}, to depict the separable capacity of statistical models. The R_{KL} is defined as

$$\mathrm{R}_{\mathrm{KL}} = \frac{d_{\mathrm{intra}}}{d_{inter}} = \frac{\mathrm{mean}(\sum \mathrm{KL}_{\mathrm{same}})}{\mathrm{mean}(\sum \mathrm{KL}_{\mathrm{diff}})} \tag{29}$$

where d_{intra} and d_{inter} denote the average KL distance of the statistical models from the same and different classes, respectively; mean( ) denotes the calculation of the mean value, $\mathrm{KL}_{\mathrm{same}}$ is the KL distance between any two frame statistical models from the same class, and the $\mathrm{KL}_{\mathrm{diff}}$ denotes the KL distance between any two frame statistical models from different class. It is clear that the smaller value of R_{KL} indicates the stronger model separability. In Table 7, we compare R_{KL} of the MTL-FA, CFA, and LCCFA models with different r values. From Table 6 we note that the R_{KL} values in the LCCFA are smaller than those in the MTL-FA and CFA, which demonstrates the effectiveness of the label constraint on the enhancement of LCCFA model's discrimination.

6. Conclusions

This study discusses a classification method based on the Bayesian probability model. For the parametric models, one critical problem is the sensitivity of their classification performance to the number of training samples. To achieve higher classification accuracy with limited training data, we develop a novel probabilistic generative model named LCCFA. In the proposed method, each dictionary atom is used as a small-scale convolution kernel and can extract the basic structures of observations, thus making it possible to represent the observed data with fewer dictionary atoms. As a result, our LCCFA model needs to estimate fewer parameters and is efficiently adapted for the classification with small training data size. Moreover, the label information is exploited to affect the model parameters' learning. With label constraint, the statistical models from different classes are more separable, thereby facilitating the further improvement of the classification performance. Experiments on benchmark datasets and measured radar HRRP data validate the effectiveness of our LCCFA model.

CRediT authorship contribution statement

Jian Chen: Methodology, Data curation, Investigation, Software, Writing - original draft. **Lan Du:** Funding acquisition, Conceptualization, Project administration, Resources, Writing - review & editing. **Yuchen Guo:** Validation, Writing - original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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